EVALUATION OF CLASSICAL AND CONTEMPORARY CLASSIFICATION METHODS

A Dissertation Report Submitted to Bharathiar University

in Partial Fulfilment of the Requirements

for the Award of the Degree of

MASTER OF SCIENCE
IN
STATISTICS

Submitted by

KARTHIK S

Reg. No. 24STAC009

Under the Guidance of

Dr. R. MUTHUKRISHNAN

PROFESSOR AND HEAD



DEPARTMENT OF STATISTICS

BHARATHIAR UNIVERSITY

COIMBATORE – 641 046

MARCH 2026

Certificate

CERTIFICATE

This is to certify that the dissertation work entitled **"Evaluation of Classical and Contemporary Classification Methods"** submitted to the Bharathiar University, Coimbatore in partial fulfilment of the requirements for the award of the degree of **Master of Science in Statistics** is a record of original work done by **KARTHIK.S (24STAC009)**, during the period (2024 - 2026) of his study in the **Department of Statistics**, **Bharathiar University**, under my supervision and guidance. This dissertation work has not formed the basis for the award of any Degree/Diploma/Association/Fellowship or similar titled to any candidate of any University/Institution.

Head of the Department
(**Dr. R. MUTHUKRISHNAN**)

Guide and Supervisor
(**Dr. R. MUTHUKRISHNAN**)

Date of Viva-Voce Examination:

External Examiner

Declaration

DECLARATION

I hereby declare that the dissertation work entitled **"Evaluation of Classical and Contemporary Classification Methods"** submitted to the Bharathiar University, Coimbatore in partial fulfilment of the requirements for the award of the degree of **Master of Science in Statistics** is a record of original work done by me under the guidance of **Dr. R. MUTHUKRISHNAN**, Professor and Head, Department of Statistics, Bharathiar University, Coimbatore. This dissertation work has not formed the basis for the award of any Degree/Diploma/Association/Fellowship or similar titled to any candidate of the University.

Signature of the Guide
(Dr. R. MUTHUKRISHNAN)

Signature of the Candidate
(**KARTHIK. S**)

Acknowledgement

ACKNOWLEDGEMENT

With immense pleasure, I express my deep sense of gratitude and profound thanks to **Dr. R. MUTHUKRISHNAN**, Professor and Head, Department of Statistics, Bharathiar University, Coimbatore for guiding me throughout this dissertation.

My wholehearted thanks to **Dr. R. VIJAYARAGHAVAN** (Retd.), Former Head and Professor and **Dr. R. JAISANKAR** (Retd.), Former Head and Professor, Department of Statistics, Bharathiar University, Coimbatore.

My profound thanks to **Dr. K. M. SAKTHIVEL**, Professor., **Dr. V. KAVIYARASU**, Associate Professor., **Dr. K. PRADEEPA VEERAKUMARI**, Associate Professor., **Dr. S. JAYALAKSHMI**, Assistant Professor and **Dr. S. GANDHIYA VENDHAN**, Assistant Professor of Department of Statistics, Bharathiar University, Coimbatore, for their valuable suggestions and encouragement.

I wish to extend my thanks to **P. JAYARAJ** and **P. SUBRAMANIAN**, Non-teaching staffs for their kind assistant for providing the necessary facilities to carry out this work.

My most heartfelt thanks to my friends both (boys & girls) and genuine thanks to **Dr. Karthika Ramakrishnan**, **Ms. G. Dheepika, Mrs. I. Mohana** and other Ph.D research scholars of the Department of Statistics for their timely help during the period of study.

It is my pride and pleasure to record my special thanks to Parents, Kavy and Friends for their help and support in the course of preparation and completion of this dissertation.

KARTHIK S

List of Abbreviations

LIST OF ABBREVIATIONS

BMI	***Body Mass Index***
BP	***Blood Pressure***
FN	***False Negative***
FP	***False Positive***
i.i.d	***Independent and Identically distributed***
IQR	***Inter Quartile Range***
LDA	***Linear Discriminant Analysis***
OOB	***out-of-bag***
QDA	***Quadratic Discriminant Analysis***
SVM	***Support Vector Machine***
TN	***True Negative***
TP	***True Positive***
WHO	***World Health Organization***

Preface

PREFACE

The dissertation embodies the work done by author under the guidance of Dr. R. Muthukrishnan. It is mainly concerned with classical and contemporary methods in the context of classification by considering various types of categorical response variable. All the analyses are carried out by using R Software.

Chapter 1 provides the Introduction to measurement scales, statistical modelling, classification techniques and R Software.

Chapter 2 presents the Methodology of various classical & contemporary classification techniques and comparing the accuracy by computing confusion matrix.

Chapter 3 deals with Experimental study of various categorical responses such as Binary, Multinomial and Ordinal under three experimental conditions, which have been carried out by using R Software.

Chapter 4 provides the Conclusions based on results. Further the scope and Suggestions also included in this chapter.

The datasets and R codes are used to carry out the experimental studies which are given in Appendix.

The references that have been made Chapter 1 to Chapter 4 are furnished in consolidated form at the end.

Contents

CONTENTS

	Title	Page No.
	Certificate	i
	Declaration	ii
	Acknowledgement	iii
	List of Abbreviations	iv
	Preface	v
Chapter 1	Introduction	01
Chapter 2	Classical and Contemporary methods	16
Chapter 3	Experimental Study	46
Chapter 4	Summary and Conclusion	55
	Appendix	58
	References	68

Introduction

CHAPTER I

INTRODUCTION

1.1 Introduction

Statistics is the science of learning from data. It deals with the methods for collecting, classifying, presenting, analyzing, and interpreting both numerical and categorical information in order to draw meaningful conclusions and support evidence-based decision making. In a world characterized by variability and uncertainty, statistics provides the tools necessary to understand patterns, measure relationships among variables, and make informed predictions from data of diverse types.

"Statistical thinking improves the quality of human life at every stage."

The need for statistical study arises from the fact that most real-life decisions must be made under conditions of uncertainty. Statistical analysis enables us to summarize complex datasets, draw reliable conclusions, test hypotheses, and make predictions about populations based on sample information. It also allows for objective comparison, risk assessment, and evidence-based decision-making. This discipline provides the tools to separate meaningful signals from random noise, ensuring that observed patterns are not merely due to chance. By quantifying variability and measuring the strength of relationships between variables, statistics transforms raw data into actionable insights.

Statistical methods provide a scientific basis for understanding variability, identifying patterns, and quantifying relationships among variables observed in real-world phenomena. These methods are broadly classified into descriptive and inferential techniques. Descriptive methods summarize and present data using measures such as averages, dispersion, tables, and graphical representations, while inferential methods enable us to make generalizations about a population based on sample data. Inferential techniques include estimation, hypothesis testing, correlation, regression analysis, classification, and survival analysis, all of which play a crucial role in scientific research.

The application of statistical methods is essential across diverse disciplines such as economics, medicine, engineering, agriculture, and social sciences. These methods help assess uncertainty, test theoretical assumptions, evaluate model performance, and ensure the reliability and validity of research findings. In contemporary research, statistical methods also form the backbone of data analysis in areas such as machine learning and predictive modelling.

This chapter presents an overview of both classical and contemporary classification techniques used to develop appropriate predictive models for categorical outcome variables measured on different scales. The fundamental concepts and preliminary background relevant to this dissertation are introduced in this chapter. In particular, the subsequent sections discuss the measurement scales of data, statistical modelling, various types of classification techniques, and the role of the R programming language in implementing these methods.

1.2 Measurement Scales

Measurement is the process of assigning numbers or labels to objects, events, or individuals according to specific rules. The scale of measurement also called the level of measurement determines what mathematical operations are permissible and what statistical analyses are appropriate for a given variable. Understanding measurement scales is fundamental to research methodology because the scale type dictates the entire analytical approach, from descriptive statistics to inferential tests and interpretation of results. Using the wrong scale for analysis can lead to false conclusions and invalid findings. It also helps to choose the right way to present the data, such as using averages, percentages, or graphs. Ultimately, proper measurement ensures that the data collected is meaningful and accurately reflects what is being studied.

The concept of measurement scales was formalized by psychologist Stevens, S. S. (1946), who proposed a four-level taxonomy comprising nominal, ordinal, interval, and ratio scales. A classification that remains the standard in research methodology across all scientific disciplines. Stevens argued that the scale of measurement determines the types of statistical operations that are meaningful for a variable, stating that "statistics cannot be more meaningful than the measurements upon which they are based".

1.2.1 Nominal scale

A nominal variable (from the Latin nomen meaning "name") is a variable that classifies observations into two or more mutually exclusive and exhaustive categories that have no numerical significance or inherent order. Nominal scales represent qualitative differences rather than quantitative differences.

Core Properties

- **Identity:** Each category has a distinct meaning and label
- **Mutual Exclusivity:** Each observation fits into exactly one category
- **Exhaustiveness:** Every observation can be classified into some category
- **No Order:** Categories cannot be meaningfully ranked or ordered
- **No Magnitude:** No category is "greater than" or "less than" another
- **No True Zero:** The concept of zero is arbitrary or meaningless.

Examples:

Types of Nominal Variables

- **Dichotomous (Binary) Nominal:** Exactly two mutually exclusive categories:
    - Heart Disease: Present / Absent
    - Gender: Male / Female
    - Survival: Alive / Deceased
- **Polytomous (Multicategory) Nominal:** Three or more unordered categories
    - **Wine Cultivars:** Class 1, Class 2, Class 3
    - Blood Type: A, B, AB, O

1.2.2 Ordinal scale

An ordinal variable (from the Latin ordo meaning "order") is a variable that classifies observations into two or more mutually exclusive and exhaustive categories that have a natural, meaningful order, but the intervals between categories are not necessarily equal or known. Ordinal scales represent both qualitative differences and quantitative ranking.

Core Properties

- **Identity:** Each category has a distinct meaning
- **Mutual Exclusivity:** Each observation fits into exactly one category
- **Exhaustiveness:** Every observation can be classified
- **Order:** Categories have a meaningful sequence or ranking
- **Magnitude:** The direction of difference is known (greater than/less than)
- **Unknown Intervals:** The distances between categories are not necessarily equal
- **No True Zero:** The concept of zero is arbitrary

Examples:

Types of Ordinal Variables

- **Ordinal variable in nature:**
    - Strongly Disagree < Disagree < Neutral < Agree < Strongly Agree
    - Never < Rarely < Sometimes < Often < Always
- **Created Ordinal Variables (from continuous)**
    - **Age Groups (from your study):** Middle-aged (39-49) < Late Middle-aged (50-59) < Young-old (60-65)
    - BMI Categories: Underweight < Normal < Overweight < Obese

1.2.3 Interval scale

An interval variable is a variable that possesses ordered categories with equal intervals between values, but no true, meaningful zero point. The zero point on an interval scale is arbitrary and does not indicate the complete absence of the attribute being measured. This means we can add and subtract values to find meaningful differences, but we cannot multiply or divide them. For example, a temperature of 20°C is not twice as hot as 10°C because 0°C does not mean "no temperature." Therefore, while we can compare distances between measurements, we cannot make statements about ratios or proportions.

Core Properties

- **Identity:** Categories are distinct
- **Order:** Meaningful sequence
- **Equal Intervals:** The difference between any two adjacent values is constant across the scale
- **No True Zero:** Zero is arbitrary and does not mean "none of the attribute"
- **Arbitrary Origin:** The zero point can be shifted without changing the meaning of differences

Examples:

- Temperature (Celsius, Fahrenheit)
- Calendar Dates (Year 2000 to 2005 = same interval as 2010 to 2015, but year 0 is arbitrary)
- IQ Scores (Difference between 100 and 110 equals 110 and 120, but zero IQ is meaningless)

1.2.4 Ratio scale

A ratio variable representing the highest and most informative level of measurement, which possesses all the properties of nominal, ordinal, and interval scales. Its defining characteristic is a non-arbitrary, absolute zero point, which indicates the complete absence of the quantity being measured. The existence of this true zero grounds the scale and permits the full range of mathematical operations addition, subtraction, multiplication, and division. Crucially, it allows for the formation and interpretation of ratios (e.g., "X is twice as large as Y"), thereby enabling both absolute and relative comparisons between values. This property is essential for analyses where proportional relationships are of primary interest.

Because the zero point is meaningful, researchers can confidently compute summary statistics such as the mean, standard deviation, and variance. Ratio scales also support advanced statistical techniques, including geometric means, coefficients of variation, and logarithmic transformations.

Core Properties

- **Identity:** Categories are distinct (nominal property)
- **Order:** Meaningful sequence (ordinal property)
- **Equal Intervals:** Differences are meaningful and equal (interval property)
- **True Zero:** Zero means "none of the attribute"
- **Ratio Meaningfulness:** Ratios between values are interpretable

Examples: Converting meters to feet: feet = 3.28 × meters

- Preserves ratios (10m/5m = 2; 32.8ft/16.4ft = 2)
- Zero remains zero (0m = 0ft)

1.3 Statistical Modelling

Statistical modelling is a fundamental framework in data analysis that aims to represent, explain, and predict real-world phenomena through mathematical relationships between variables. By combining probability theory with empirical data, statistical models enable us to quantify uncertainty, test scientific hypotheses, and draw meaningful inferences from observed data. In many disciplines statistical modelling serves as a crucial tool for evidence-based decision making.

Among the various approaches to statistical modelling, regression analysis occupies a central role due to its ability to describe the relationship between a response variable and one or more explanatory variables. Regression models are particularly valued for their interpretability and flexibility, allowing statisticians to assess the direction, magnitude, and significance of covariate effects while controlling for confounding factors.

In the 1880s, Sir Francis Galton wanted to understand how traits like height are passed from parents to children. He collected data on many families and noticed something strange: very tall parents usually had children who were tall, but not as tall as them. The same happened for very short parents; their children were short, but not as short. It seemed like the children's heights were moving back, or "regressing," toward the average height of everyone else. Galton called this idea "regression towards mediocrity," which is why we still use the name "regression" today. This represented a significant development because it turned a simple math tool, originally created to track planets, into a way to understand biology and life.

Regression techniques can be broadly classified based on the nature of the outcome variable. When the response variable is numerical or continuous, models such as linear regression and generalized linear models are commonly employed to estimate mean responses and identify linear or nonlinear trends. Conversely, when the outcome variable is categorical, regression frameworks such as logistic regression, multinomial regression, and ordinal regression are used to model probabilities associated with different categories. These models extend classical regression concepts by incorporating suitable link functions and distributional assumptions tailored to categorical data.

The unifying strength of regression-based statistical modelling lies in its capacity to handle both numerical and categorical outcomes within a coherent theoretical framework. This adaptability makes regression an indispensable analytical tool for modern statistical research, enabling robust prediction, classification, and inference across a wide range of applications. While this foundation is well-established, the performance of different regression-based and machine learning classification models can vary significantly depending on the data structure, particularly the nature of the categorical outcome. This variability motivates the comparative investigation undertaken in this dissertation.

1.4 Classification

Classification is a fundamental problem in statistical learning that involves assigning observations to one of several predefined categories based on a set of explanatory variables. Unlike regression models that focus on predicting numerical outcomes, classification methods are specifically designed to handle categorical response variables, making them essential in studies where the outcome represents qualitative states, group membership, or decision classes.

From a statistical perspective, classification is concerned with estimating the probability that an observation belongs to a particular class and constructing decision rules that minimize misclassification. These methods rely on both parametric and non-parametric modelling frameworks, enabling flexibility in capturing complex relationships between predictors and categorical outcomes.

In this dissertation, the primary focus is on supervised statistical classification for categorical outcomes. The study systematically examines the theoretical foundations, underlying assumptions, and comparative predictive performance of a suite of classification models. Through this investigation, the dissertation aims to provide practical guidance on model selection for researchers dealing with categorical data, highlighting the trade-offs between interpretability and predictive accuracy.

1.5 Classification Techniques

Classification techniques are an important part of predictive modeling, designed to assign observations to predefined categories based on a set of input variables. These methods vary widely in their underlying principles, assumptions, and computational approaches, ranging from classical statistical models rooted in probability theory to modern machine learning algorithms capable of capturing complex patterns in data. The techniques discussed in this section Binary, Multinomial, and Ordinal Logistic Regression; Linear and Quadratic Discriminant Analysis; Random Forest and its ordinal variant; Support Vector Machines; and XGBoost, these methods represent a comprehensive toolkit for addressing classification problems across diverse data structures.

Each method offers distinct advantages depending on the nature of the outcome variable (binary, nominal, or ordinal) and the distribution of the predictor. The following sections provide a concise explanation of each technique, including its core concept and historical context.

1.6 Classical Classification Methods

Classical statistical methods form the foundation for developing classification models when the response variable is categorical. Some of the widely used approaches include Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and logistic regression models. LDA and QDA classify observations by analyzing the distributional characteristics of predictor variables across different groups, while logistic regression models estimate the probability of class membership using the logistic function.

These classical techniques are based on certain statistical assumptions, such as distributional properties of the predictors and relationships between variables. Depending on the nature of the response variable, logistic regression can be applied as binary, multinomial, or ordinal logistic regression. Despite their assumptions, these methods remain widely used because of their interpretability and strong theoretical foundation in statistical modelling.

1.6.1 Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis was introduced by the eminent statistician Sir Fisher, R. A. (1936). Working at Rothamsted Experimental Station, Fisher sought to address a practical classification problem: distinguishing among three species of iris flowers (Iris setosa, Iris versicolor, and Iris virginica) using measurements of sepal length, sepal width, petal length, and petal width.

Fisher's innovation was to derive a linear combination of features that maximally separates classes while minimizing within-class variation. Technically, LDA projects the data onto a lower-dimensional space by maximizing the ratio of between-class variance to within-class variance, thereby creating a discriminant function that best separates the groups. A new observation is then classified based on its proximity to each class's mean in this transformed space.

Despite its historical significance and continued application, LDA's performance relative to modern classifiers remains underexplored in several contexts. Specifically, the extent to which violations of its normality and homoscedasticity assumptions degrade predictive accuracy compared to more flexible algorithms warrants systematic investigation. Furthermore, while LDA is theoretically optimal under its assumptions, empirical guidance on when its interpretability advantages outweigh potential performance losses remains limited in the literature.

1.6.2 Quadratic Discriminant Analysis (QDA)

Quadratic Discriminant Analysis emerged as a natural extension of Fisher's LDA, developed by Smith, C. A. B. (1947). It relaxes a core assumption of LDA that all classes have the same covariance matrix. Instead, QDA estimates and employs a separate covariance matrix for each class. This added flexibility allows it to create more complex, non-linear (quadratic) decision boundaries, making it better suited for datasets where the spread of data points differs significantly between categories. However, this flexibility comes at a cost, as estimating more parameters requires larger datasets to avoid overfitting, keeping it firmly rooted in the parametric statistical tradition alongside LDA.

1.6.3 Binary Logistic Regression

In the middle of the 20th century, statisticians ran into a problem with the traditional regression tool created by Galton. Their old method worked well for predicting things like height or weight, but it failed when trying to predict simple yes-or-no outcomes, like whether someone would get a disease or whether a customer would buy a product. Scientists needed a way to handle these kinds of questions, where the answer falls into one of two categories.

Unlike linear regression, which models an unbounded continuous outcome, logistic regression employs the sigmoid (logistic) function to constrain predicted values within the interval [0, 1]. This property renders it particularly suitable for modeling probabilities associated with binary outcomes. The development of logistic regression represented a significant methodological advancement, enabling researchers to estimate the probability of event occurrence across diverse domains.

While its mathematical foundations lie in the 19th-century work on the logistic function for population growth, Binary Logistic Regression became a mainstay of statistical analysis in the 1950s and 1960s, particularly in fields like biostatistics and epidemiology. A major contribution to the development and application of this method was made by Cox, D. R. (1958), who formalized logistic regression models for the analysis of binary response data. Unlike LDA, which models class boundaries, this method models the probability of a binary outcome (e.g., alive/dead, pass/fail). It applies the logistic (sigmoid) function to a linear combination of predictors, transforming the output into a probability value between 0 and 1.

Binary logistic regression is used when the response variable has two possible categories (e.g., success/failure, presence/absence). The model estimates the log-odds of the probability of an event occurring as a linear function of the predictors. Regression coefficients are interpreted in terms of odds ratios, making the model particularly attractive for inferential purposes.

Its primary strength lies in its interpretability, providing direct estimates of odds ratios. However, it assumes a linear relationship between the log-odds and the predictors and can be less effective if complex, non-linear interactions are present.

1.6.4 Multinomial Logistic Regression

As the field of categorical data analysis advanced in the 1960s and 1970s, heavily influenced by statisticians like Theil, H. (1969) & Nelder, J. A. (1989). Binary Logistic Regression was logically extended to handle dependent variables with more than two nominal (unordered) categories.

Multinomial Logistic Regression is this direct generalization. It works by effectively running multiple binary logistic regressions simultaneously, comparing each category of the outcome to a chosen baseline category. For instance, predicting whether a patient has a cold, flu, or allergies would involve models comparing cold vs. flu, and cold vs. allergies. This provides a cohesive framework for understanding how predictors influence the odds of being in one category versus another, without imposing any order on the categories themselves.

1.6.5 Ordinal Logistic Regression

The challenge of modelling response variables that have a natural and meaningful ordering, such as disease severity levels (none, mild, severe), bond ratings (AAA, AA, A), or satisfaction ratings, was formally addressed by McCullagh, P. (1980). He introduced the most common and rigorous implementation, the proportional odds model. The core concept is to model the cumulative probability (the probability of being at or below a particular category) rather than the probability of individual categories. This structure allows the model to make efficient use of the ordering information inherent in the response variable while maintaining statistical validity and interpretability.

A key assumption of the model is that the effect of the predictor variables is constant across the different thresholds of the ordinal scale. In other words, the factors that push someone from "mild" to "severe" pain are assumed to be the same as those that push them from "none" to "mild." This provided a powerful and elegant tool for a vast range of social, medical, and economic data that was previously difficult to analyze correctly.

1.7 Contemporary Classification Methods

Contemporary classification methods have emerged to address the limitations of classical statistical models, particularly when dealing with nonlinear relationships, high-dimensional data, and complex interactions among variables. Unlike classical approaches, which often rely on strict assumptions such as normality, linearity, and independence of predictors, these contemporary techniques are largely assumption-free or make minimal assumptions about the underlying data distribution.

They emphasize predictive accuracy and generalization performance, often at the expense of model interpretability, through the use of advanced concepts such as regularization, ensemble learning, and kernel-based transformations. Due to their robustness, flexibility, and ability to capture complex patterns, these methods have been widely adopted across diverse domains, including bioinformatics, finance, image analysis, and machine learning, where they consistently achieve state-of-the-art performance in challenging prediction tasks.

1.7.1 Support Vector Machines (SVM)

The theoretical foundations of Support Vector Machines were established by Vladimir Vapnik and Alexey Chervonenkis in the 1960s. The method later gained widespread recognition after the publication of the research paper by Cortes, C., & Vapnik, V. (1995). SVM is a powerful classification algorithm with an elegant concept: it finds the optimal decision boundary (hyperplane) that not only separates classes but does so by maximizing the margin, the distance between the boundary and the nearest data points from each class. These critical data points are the support vectors.

Crucially, the "kernel trick" allows SVM to create non-linear decision boundaries by implicitly mapping the data into a higher-dimensional space, all without becoming computationally overwhelmed. This combination of strong theoretical foundations and excellent performance made it a leading algorithm in the machine learning revolution of the 1990s.

1.7.2 Random Forest

The Random Forest algorithm was formally introduced by Breiman, L. (2001). The method builds upon his earlier work on bagging and incorporates ideas from the random subspace method to improve prediction accuracy and model stability.

Random forests are a combination of tree predictors such that each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest. The generalization error of a forest of tree classifiers depends on the strength of the individual trees in the forest and the correlation between them. Using a random selection of features to split each node yields error rates that compare favorably to Adaboost (Freund and Schapire [1996]), but are more robust with respect to noise. Internal estimates monitor error, strength, and correlation and these are used to show the response to increasing the number of features used in the splitting. Internal estimates are also used to measure variable importance.

Its key strength is its high predictive accuracy and ability to model complex, non-linear relationships with minimal assumptions. This robustness, however, comes at the cost of interpretability, as the ensemble of trees is a "black box" compared to a single logistic regression model.

1.7.3 XGBoost (Extreme Gradient Boosting)

XGBoost was developed by Chen, T., & Guestrin C. (2016). Rather than introducing a completely new algorithm, XGBoost represents a significant engineering advancement that enhanced and popularized the existing concept of gradient boosting, improving its efficiency, scalability, and predictive performance. Boosting, pioneered in the 1990s with algorithms like AdaBoost and formalized by Jerome Friedman in his 2001 paper on "Gradient Boosting Machines," builds models sequentially, where each new tree corrects the errors of the previous one. XGBoost's effectiveness stems from its highly optimized and scalable implementation of this idea. It incorporates regularization to prevent overfitting, efficiently handles missing values, and is designed for exceptional computational efficiency. Its dominance in machine learning competitions and industry applications.

1.7.4 Ordinal Forest

Reflecting the ongoing trend of tailoring powerful generic algorithms for specific data structures, Ordinal Forest was developed in the 2010s, with a prominent method proposed by Hornung, R. (2020). It directly adapts Breiman's Random Forest for ordinal regression problems. The core concept involves modifying the algorithm to respect the order of the outcome variable during both the tree-building and prediction phases. This can be achieved by using a splitting criterion for the nodes that is specifically designed for ordered categories, or by aggregating the predictions from individual trees in a way that reflects the ordinal scale (e.g., by averaging the predicted class indices). This specialization allows the model to leverage the power of random forests while making predictions that are intrinsically consistent with an ordered outcome.

1.8 Software

R is a powerful open-source programming language and software environment specifically designed for statistical computing, data analysis, and graphical visualization. The story of R begins in the early 1990s at the University of Auckland in New Zealand, where statisticians Ross Ihaka and Robert Gentleman started developing the language to teach introductory statistics courses. Frustrated by the lack of practical, accessible software for their needs, they created R as an implementation of the S programming language, which had been developed at Bell Laboratories.

The name "R" reflects both the shared first initial of its two creators and a nod to its predecessor, S. In a pivotal moment for its future, R became part of the GNU Project in 1997, ensuring it would remain free and open-source software. With the release of version 1.0 in February 2000, R was solidified as a stable, powerful platform.

Since then, it has grown from a classroom project into one of the world's most widely used tools for data science, supported by a vast community that has contributed over 23,000 packages (as of March 2026) to extend its capabilities. Its popularity stems from this extensive collection of packages, which provide specialized functions for basic statistics to advanced machine learning. While R has a steeper learning curve than some point-and-click statistical software, its flexibility, reproducibility, and the active support of its global user community make it an indispensable tool for transforming raw data into meaningful insights through sophisticated analysis and publication-quality graphics.

Classical & Contemporary Methods

CHAPTER II

CLASSICAL AND CONTEMPORARY METHODS

2.1 Introduction

The prediction of categorical outcomes is an important problem in many fields such as medicine, social sciences, and marketing. In classification problems, observations are placed into different groups based on a set of predictor variables. The type of categorical outcome such as binary, nominal, or ordinal plays an important role in choosing the appropriate statistical method and in interpreting the results.

Categorical outcomes can appear in three main forms. Binary outcomes have two categories, such as success/failure or yes/no. Nominal outcomes have three or more categories without any natural order, such as different product brands or types of diseases. Ordinal outcomes also have multiple categories, but they follow a natural order, such as low, medium, and high, or levels of disease severity.

This chapter provides the basic framework for the study by comparing different classification methods for different types of categorical outcomes. It presents the research design used in the study and discusses various classical and contemporary classification techniques along with their model structure, underlying assumptions, advantages, and limitations. To evaluate and compare the performance of the models, confusion matrices and accuracy metrics are used. These measures help assess the efficiency of each classification method by examining how well the models correctly classify the observations.

2.2 Research Design

The large number of classification algorithms provides researchers with many options, but it also makes method selection more challenging. The performance of these methods can vary depending on the type of outcome variable, the structure of the data, and the objective of the study. Traditional parametric methods such as logistic regression and discriminant analysis are easy to interpret and useful for statistical inference.

In contrast, modern machine learning methods like Random Forest, XGBoost, and Support Vector Machines can capture complex and non-linear relationships and often provide better predictive accuracy. Therefore, choosing a suitable classification method requires balancing interpretability, predictive performance, and the assumptions of the model.

To compare the performance of these methods, confusion matrices and accuracy measures are used. The confusion matrix summarizes correct and incorrect classifications for each category and helps calculate performance measures such as sensitivity, specificity, and positive predictive value. Overall classification accuracy is used as the main indicator of model performance. This framework allows a systematic comparison of classification methods and provides a basis for selecting suitable models in the later empirical chapters of this study.

2.3 Classification Methods

Classification techniques are predictive modelling methods used to assign observations into predefined categories based on input variables. These methods range from traditional statistical models to modern machine learning algorithms, each with different principles and assumptions. The key techniques covered include Binary, Multinomial, and Ordinal Logistic Regression; Linear and Quadratic Discriminant Analysis; Random Forest; Support Vector Machines; and XGBoost. Together, they provide a comprehensive toolkit for solving classification problems across various types of data.

2.3.1 Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis was introduced by the eminent statistician Sir Fisher, R. A. (1936). It is a classical parametric classification method that assumes multivariate normality with equal covariance matrices across classes. It classifies observations by maximizing linear discriminant functions derived from Bayes' theorem, resulting in linear decision boundaries. LDA is computationally efficient, interpretable, and optimal under its assumptions, making it a widely used traditional method for supervised classification.

Mathematical formulation

Let $Y \in \{1,2,...,K\}$ be a categorical response variable, $X = (X_1, X_2, ..., X_p)^T$ be a vector of predictors and LDA models the class-conditional distribution of predictors.

For class *k* the multivariate normal density is:

$$\boldsymbol{f_k(x)} = \frac{1}{(2\pi)^{p/2}\,|\boldsymbol{\Sigma}|^{1/2}}\ \mathbf{exp}\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu_k})^{\mathbf{T}}\,\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu_k})\right] \qquad (2.1)$$

Where μ_k is mean vector of class *k* and Σ is common covariance matrix

Using Bayes' theorem and simplifying the log-posterior probability yields the linear discriminant function model

$$\boldsymbol{\delta_k(x)} = \boldsymbol{x^T\Sigma^{-1}\mu_k} - \frac{1}{2}\boldsymbol{\mu_k^T\,\Sigma^{-1}\mu_k} + \mathbf{log}\ \boldsymbol{\pi_k} \qquad (2.2)$$

Where $x^T\Sigma^{-1}\mu_k$ measures similarity between observation and class means, $\mu_k^T\,\Sigma^{-1}\mu_k$ penalty for class location and $\log \pi_k$ incorporates prior belief about class frequency.

Model Assumptions

- Multivariate normality:

$$\mathbf{X}\ |\ (\mathbf{Y}=\mathbf{k})\ \sim\ \boldsymbol{N}\mathbf{p}(\boldsymbol{\mu_k}, \boldsymbol{\Sigma})$$

- Equal covariance matrices across classes:

$$\boldsymbol{\Sigma_1} = \boldsymbol{\Sigma_2} = \cdots\cdots = \boldsymbol{\Sigma_k} = \boldsymbol{\Sigma}$$

- Independence of observations
- Predictors are continuous
- No perfect multicollinearity

Advantages

- **Classification and Dimensionality Reduction -** Linear discriminant analysis seeks to find the direction which maximizes between-class variation and minimizes the within-class variation.
- **Foundation in Bayes Theorem -** A simple application of Bayes theorem gives us the posterior class probabilities. Linear discriminant analysis (LDA) arises in the special case when we assume that the classes have a common covariance matrix.
- **Proven Effectiveness in Practice -** LDA perform well on an amazingly large and diverse set of classification tasks.

Limitations

- **Sensitivity to Outliers -** Since LDA relies on sample means and pooled covariance matrices, it is highly sensitive to outliers. A single extreme observation can disproportionately influence the estimated discriminant functions.
- **Poor Handling of Categorical Predictors -** LDA assumes continuous predictors that follow a multivariate normal distribution. Binary or categorical predictors violate this assumption and can lead to poor performance.
- **Susceptibility to Multicollinearity -** Severe multicollinearity among predictor variables can cause unstable estimates of the discriminant coefficients, similar to its effects in multiple regression.
- **Assumes Independent Observations -** Repeated measures or hierarchical data structures cannot be properly modelled with standard LDA without modification.
- **Difficulty with Heterogeneous Groups -** LDA assumes that each group is homogeneous and follows a single multivariate normal distribution. When groups contain distinct subgroups or mixtures, LDA's performance suffers.

Linear Discriminant Analysis (LDA) is preferable when the primary objective is dimensionality reduction or classification with the additional goal of understanding how predictors differentiate between groups, particularly when the assumptions of multivariate normality and equal covariance matrices across groups are reasonably satisfied.

LDA provides more efficient and stable estimates than logistic regression. Additionally, it explicitly identifies linear combinations of predictors that maximize between-group variance relative to within-group variance.

2.3.2 Quadratic Discriminant Analysis (QDA)

Quadratic Discriminant Analysis (QDA), which emerged as a natural extension of Fisher's Linear Discriminant Analysis Smith, C. A. B. (1947). It is a parametric classification method that allows class-specific covariance matrices. By modeling predictors as multivariate normal distributions within each class and applying Bayes' theorem, QDA derives quadratic discriminant functions that produce nonlinear decision boundaries capable of capturing more complex data patterns.

Let $Y \in \{1,2,\ldots,K\}$ be a categorical response variable and $X = (X_1, X_2, \ldots, X_p)^T$ be a vector of predictors. QDA models the class-conditional distribution of predictors. For class k the multivariate normal density is:

$$\boldsymbol{f_k(x)} = \frac{1}{(2\pi)^{p/2}\,|\boldsymbol{\Sigma_k}|^{1/2}}\ \mathbf{exp}\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu_k})^{\mathbf{T}}\,\boldsymbol{\Sigma_k^{-1}}(\mathbf{x}-\boldsymbol{\mu_k})\right] \tag{2.3}$$

Where, μ_k is mean vector of class k and Σ_k is covariance matrix of class k

Using Bayes' theorem and simplifying the log-posterior probability yields the quadratic discriminant function model

$$\boldsymbol{\delta_k(x)} = -\frac{1}{2}\mathbf{log}\,|\,\boldsymbol{\Sigma_k}\,| - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu_k})^{\mathbf{T}}\,\boldsymbol{\Sigma_k^{-1}}(\mathbf{x}-\boldsymbol{\mu_k}) + \mathbf{log}\,\boldsymbol{\pi_k} \tag{2.4}$$

Where $(x-\mu_k)^T\,\Sigma_k^{-1}(x-\mu_k)$ is Mahalanobis distance from class mean, $\log|\,\Sigma_k\,|$ is penalty for class dispersion and $\log \pi_k$ is incorporates prior class information.

Model Assumptions

- Multivariate normality:

$$\mathbf{X} \mid (\mathbf{Y} = \mathbf{k}) \sim \boldsymbol{N}\mathbf{p}(\boldsymbol{\mu_k}, \boldsymbol{\Sigma}\mathbf{k}), \mathbf{k} = \mathbf{1}, \mathbf{2}, \ldots, \mathbf{K}$$

- Class-specific covariance matrices:

$$\boldsymbol{\Sigma}_{\mathbf{k}} \neq \boldsymbol{\Sigma}_{\mathbf{j}} \textbf{ for } \mathbf{k} \neq \mathbf{j}$$

- Independence of observations
- Predictors are continuous
- No perfect multicollinearity within classes

Advantages

- **Greater Flexibility with Decision Boundaries -** When the data present nonlinear patterns, linear decision boundaries may not separate different classes effectively. QDA, by using quadratic decision boundaries, can better handle these nonlinear problems.
- **No Assumption of Equal Covariance Matrices -** QDA relaxes the assumption made by LDA-based methods that the covariance matrix for each class is identical.
- **Better Fit for Complex Data Structures -** QDA, because it allows for more flexibility for the covariance matrix, tends to fit the data better than LDA.
- **Lower Bias and Better Performance with Larger Datasets -** QDA is better suited for large datasets, as it tends to have a lower bias and higher variance. On the other hand, LDA is suitable for smaller datasets that have a lower bias and a higher variance.
- **Handles Heterogeneous Class Distributions -** In the QDA framework, the class conditional distribution is assumed to be Gaussian, however, with an allowance for different covariance matrices. In such cases, a more complex quadratic boundary can be formed.

Limitations

- **High Dimensionality Problem -** The QDA fails to classify modern high-dimensional datasets consisting of a large number of feature vectors with much smaller sample sizes (n).
- **Requires Large Sample Size -** The cost of the flexibility is to estimate more parameters of the QDA model, which requires a large sample size.
- **Extreme Sensitivity to Outliers -** The classification rule in QDA is still based on the sample mean vector and the sample dispersion matrix of a training set, which are extremely non-robust under data contamination.
- **Risk of Overfitting -** The quadratic rule is used when separate group covariance matrices are used. These covariance matrices do not need to be as homogenous. However, the quadratic rule's external generalizability is not as accurate.
- **Computational Complexity -** To estimate the parameters required in quadratic discrimination more computation and data is required than in the case of linear discrimination.

Quadratic Discriminant Analysis (QDA) is preferable when the assumption of equal covariance matrices across groups required by Linear Discriminant Analysis is violated, meaning that the variance-covariance structure differs substantially between classes.

QDA is particularly advantageous when dealing with datasets where groups exhibit distinct spreads or orientations, as its quadratic decision boundary can better capture complex, nonlinear separations between categories. It performs well when the sample size is sufficiently large to reliably estimate separate covariance matrices for each class, as the increased flexibility comes at the cost of requiring more parameters.

QDA is also preferred when the relationships between predictors within each class are nonlinear on the log-odds scale, and when the primary goal is maximizing classification accuracy rather than model interpretability, as the quadratic boundaries are more difficult to summarize than linear ones.

2.3.3 Binary Logistic Regression

Binary logistic regression is a foundational parametric technique for modeling a dichotomous outcome developed by Cox, D. R. (1958). Rather than predicting the class directly, it estimates the probability that an observation belongs to a specific category (e.g., Y=1) using the logistic (sigmoid) function. This function maps any real-valued linear combination of predictors to a probability bounded between 0 and 1, producing the characteristic S-shaped curve. The model is formally expressed as:

Let Y be a categorical response variable, p is P (Y=1 | X) probability of success with X_1,X_2,…,X_k independent variables

Logistic (Sigmoid) Function

The logistic function, originally introduced by Pierre François Verhulst in the context of population growth studies, logistic regression applies the logit link function via the sigmoid function to a linear predictor: p, Logistic regression predicts the outcome:

$$\boldsymbol{p} = \boldsymbol{P}(\boldsymbol{Y} = \mathbf{1} \mid \boldsymbol{X}) = \frac{\mathbf{1}}{\mathbf{1}+ e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots\ldots + \beta_n x_n)}} \quad (2.5)$$

Where, p is the predicted probability of the outcome Y = 1, 1 – p describes Probability of the outcome Y = 0, $\boldsymbol{\beta}_0$ is Intercept(bias), $\boldsymbol{\beta}_0$, $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, ...,$\boldsymbol{\beta}_n$ is Coefficient of the independent variables $\boldsymbol{x}_1$, $\boldsymbol{x}_2$,.....,$\boldsymbol{x}_n$. This function maps any real value to the interval (0,1).

Logit Transformation

The logit is the natural logarithm of the odds:

$$\boldsymbol{logit}(\boldsymbol{p}) = \boldsymbol{log}\left(\frac{\boldsymbol{p}}{\mathbf{1}-\boldsymbol{p}}\right)$$

Model:

Let $p = P(Y=1 \mid X)$

$$\boldsymbol{log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k} \tag{2.6}$$

Or equivalently,

$$\boldsymbol{p} = \frac{\boldsymbol{e^{-(\beta^0 + \beta^1 X^1 + \beta^2 X^2 + \ldots\ldots + \beta_k X_k)}}}{\boldsymbol{1 + e^{-(\beta^0 + \beta^1 X^1 + \beta^2 X^2 + \ldots\ldots + \beta_k X_k)}}} \tag{2.7}$$

Where, p is $P(Y=1|X_1,...,X_k)$, $\boldsymbol{\beta_0}$ is intercept term, $\boldsymbol{\beta_k}$ is change in log-odds of success for a one-unit increase in X_k, holding other variables constant and X_k is k-th independent variable.

Assumptions

- **Binary Outcome Variable** - The outcome variable in logistic regression must be dichotomous, meaning it takes only two values representing the occurrence or non-occurrence of an event.
- **Independence of Observations** - The observations must be independent of each other. This means that repeated measurements or matched data should not be analyzed using standard logistic regression without modifications
- **Absence of Multicollinearity** - The independent variables should not be too highly correlated with one another, this multicollinearity can cause unstable estimates and inflated standard errors in logistic regression, similar to its effects in linear regression.
- **Linearity in the Logit** - Unlike ordinary linear regression, logistic regression does not require a linear relationship between the predictors and the outcome variable. However, it does assume linearity between each continuous predictor and the logit of the outcome. This assumption can be tested by including interaction terms between the predictor and its logarithm.
- **No Strong Influential Outliers** - Logistic regression is sensitive to extreme observations. Influential cases can disproportionately affect parameter estimates and should be examined using diagnostic statistics such as Cook's distance, leverage values, and standardized residuals.

Advantages

- **No Normality Assumption Required -** Unlike discriminant analysis, logistic regression does not require the assumption that the independent variables are normally distributed.
- **Interpretable Results as Odds Ratios -** One of the greatest strengths of logistic regression is the interpretability of its results.
- **Flexibility with Predictor Types -** Logistic regression accommodates both continuous and categorical predictor variables without requiring special adjustments. Categorical variables can be incorporated through dummy coding or other contrast methods.
- **No Homoscedasticity Assumption -** The variance in logistic regression is inherently heteroscedastic because it depends on the predicted probability.
- **Handles Nonlinear Relationships** - Although the logit is linear in the parameters, the relationship between the predictors and the probability of the outcome is nonlinear due to the logistic transformation. This allows logistic regression to model S-shaped relationships that linear probability models cannot capture.

Limitations

- **Linearity in the Logit Assumption** - Logistic regression assumes a linear relationship between each continuous predictor and the logit of the outcome. When this assumption is violated, the model may produce biased estimates and poor fit.
- **Sensitivity to Outliers -** Logistic regression is sensitive to extreme observations. A small number of influential cases can substantially alter parameter estimates and affect model conclusions.
- **Multicollinearity Problems -** Severe multicollinearity among predictors can inflate standard errors, making coefficients unstable and difficult to interpret. In extreme cases, it can prevent model convergence.
- **Difficulty with Complex Relationships** - While logistic regression handles simple nonlinear relationships through the logistic transformation, it may struggle with highly complex, interactive patterns that more flexible machine learning methods can capture.
- **Limited to Binary Outcomes -** When the outcome has more than two categories without natural ordering, multinomial logistic regression must be used, which estimates multiple sets of coefficients and can be more difficult to interpret.

Logistic regression is particularly preferable when the research objective involves binary classification with a focus on interpretability rather than pure predictive accuracy. It is especially suitable when the goal is to understand the direction and magnitude of relationships between predictors and a dichotomous outcome, as the exponentiated coefficients provide directly interpretable odds ratios.

2.3.4 Multinomial Logistic Regression

Multinomial logistic regression is a classification technique designed for situations where the dependent variable is nominal in nature (categorical with three or more categories that have no natural order) it was evolved by Theil, H. (1969) & Nelder, J. A. (1989). It predicts the probability of an observation belonging to each category by performing simultaneous pairwise comparisons between all categories and a chosen baseline reference category, subsequently transforming these comparisons into a probability vector that sums to unity.

Model

Assume response variable Y has J categories. Choose one category as the reference category (say category J) for $j = 1,2,...,J-1$

$$\boldsymbol{log}\left(\frac{\boldsymbol{P(Y=j)}}{\boldsymbol{P(Y=J)}}\right) = \boldsymbol{\beta_{0j}} + \boldsymbol{\beta_{1j}X_1} + \cdots + \boldsymbol{\beta_{kj}X_k} \tag{2.8}$$

Each category has its own set of regression coefficients**.**

Probability Form

$$\boldsymbol{P}(\mathbf{Y} = \mathbf{j}) = \frac{\boldsymbol{e}^{\left(\beta_{0j} + \beta_{1j}X_1 + \beta_{2j}X_2 + \cdots\cdots + \beta_{kj}X_k\right)}}{\mathbf{1} + \sum_{j=1}^{J-1} \boldsymbol{e}^{\left(\beta_{0j} + \beta_{1j}X_1 + \beta_{2j}X_2 + \cdots\cdots + \beta_{kj}X_k\right)}} \tag{2.9}$$

Where, β_j is parameter vector for category j and Each category is compared relative to the reference category with $\sum_{j=1}^{J-1} \boldsymbol{p_j} = \mathbf{1}$

Assumptions

- **Independence of Observations -** A fundamental assumption of maximum likelihood estimation is that the responses from different subjects are independent.
- **Absence of Multicollinearity** - Severe multicollinearity may produce implausibly large parameter estimates or standard errors, signalling that the data cannot reliably distinguish the unique effects of correlated predictors.
- **Linearity in the Logit for Continuous Predictors** - Multinomial logistic regression assumes a linear relationship between each continuous predictor and the logit of the outcome. When this assumption is violated, the model may produce biased estimates and poor fit.
- **Absence of Outliers and Influential Cases -** Multinomial logistic regression is sensitive to outliers and influential observations that can disproportionately affect parameter estimates and standard errors.
- **No Measurement Error in Predictors -** When predictors are measured with substantial error, to resolve it consider latent variable modelling approaches or errors-in-variables corrections to obtain unbiased estimates of predictor effects.

Advantages

- **Nominal Outcomes with Multiple Categories -** The primary advantage of multinomial logistic regression is its ability to model nominal outcome variables with three or more categories without imposing any ordering on the response categories.
- **No Normality or Homoscedasticity Assumptions for Predictors -** Multinomial logistic regression makes no assumptions about the distribution of the predictor variables. Predictors need not be normally distributed, linearly related to the outcome, or exhibit equal variances across groups.
- **Accommodates Mixed Predictor Types** - Multinomial logistic regression seamlessly accommodates continuous, ordinal, and nominal predictor variables within the same model. Categorical predictors can be entered through dummy coding.
- **Enables Probability Predictions for Individual Cases -** A significant practical advantage of multinomial logistic regression is its ability to generate predicted probabilities for each outcome category for individual cases.

Limitations

- **Assumption of Linear Relationship in the Logit -** Multinomial logistic regression assumes a linear relationship between each continuous predictor and the logit of the outcome. When this assumption is violated, the model may produce biased estimates and poor fit.
- **Handling High-Dimensional Data -** Multinomial logistic regression becomes increasingly unstable as the number of predictors grows relative to the sample size.
- **Sensitivity to Outliers and Influential Cases -** Like all maximum likelihood procedures, multinomial logistic regression is sensitive to outliers and influential observations that can disproportionately affect parameter estimates, standard errors, and model fit.
- **Susceptibility to Multicollinearity -** High correlations among predictor variables can inflate standard errors, produce unstable coefficient estimates, and obscure the unique contributions of individual predictors.
- **No Built-in Handling of Ordered Categories -** Multinomial logistic regression treats all categories as nominal, ignoring any natural ordering that may exist among outcome categories.

Multinomial logistic regression is preferable when the outcome variable is categorical with three or more categories that have no natural ordering (nominal outcome). This technique is specifically designed to handle situations where the dependent variable consists of multiple unordered categories, for example., predicting mode of transportation (car/bus/train).

Unlike binary logistic regression which models a single event versus non-event, multinomial logistic regression simultaneously estimates multiple binary comparisons, typically by selecting one category as the reference and modeling the log-odds of each other category relative to that reference.

2.3.5 Ordinal Logistic Regression

Ordinal logistic regression (also known as the ordinal logit, ordered polytomous logit, or proportional odds model) was formally introduced by McCullagh, P. (1980). It is a versatile method for modelling relationships between an ordinal dependent variable and multiple independent variables. It focuses on cumulative probabilities that is, the probability that the dependent variable is less than or equal to a particular value and the odds and odds ratios based on these cumulative probabilities.

This approach is suitable when the dependent variable is multi-categorical with ordered categories (e.g., quality of life rated as "poor," "fair," "good," "excellent") and the statistical model should not assume equal intervals between adjacent categories.

Proportional Odds Model

Let response variable have J ordered categories, $Y \in \{1,2,\ldots,J\}$ with natural ordering where $J \geq 3$

Cumulative probabilities defines,

$$\mathbf{P(Y \leq j),\ \ j = 1, 2, \ldots, J - 1}$$

Model

$$\boldsymbol{log\left(\frac{P(Y \leq j)}{P(Y > J)}\right) = \alpha_j - (\beta_1 X_1 + \cdots + \beta_k X_k)} \qquad (2.10)$$

Where, α_j is threshold (cut-point) parameters, β is common slope parameters across categories

Interpretation via Odds Ratios

For all logistic models:

$$\text{Odds Ratio} = \boldsymbol{e^{\beta_j}}$$

If $e^{\beta_j} > 1$ is increases probability of higher outcome and $e^{\beta_j} < 1$ is decreases probability

Assumptions

- **Proportional Odds Assumption -** The defining assumption of the ordinal logistic regression model is the proportional odds assumption. That is coefficients (slopes) for the predictors are constant across all cumulative splits of the ordinal outcome, and only the intercepts (thresholds) differ.
- **Absence of Multicollinearity -** While ordinal logistic regression does not require normally distributed predictors, it is highly sensitive to multicollinearity high correlations among the independent variables.
- **No Perfect Separation -** Ordinal logistic regression models require that no perfect separation exists in the data. Perfect separation occurs when the outcome variable is completely predicted by one or a combination of independent variables, leading to infinitely large parameter estimates and a failure of the model to converge.
- **Large Sample Size Requirement -** Adequate sample size is essential for reliable parameter estimation and hypothesis testing in ordinal logistic regression.

Advantages

- **Preservation of Ordinal Information -** A fundamental advantage of ordinal logistic regression is its ability to preserve and utilize the natural ordering of the outcome categories.
- **Flexibility with Multiple Link Functions -** The availability of multiple link functions makes ordinal models adaptable to diverse data structures.
- **Accommodation of Assessed Ordinal Variables -** These flexible ordinal models allow for testing whether covariates distinguish between specific outcome levels and can accommodate situations where different linear combinations are needed to discriminate between different categories, providing greater versatility than proportional odds models alone.
- **Enhanced Convergence Properties -** The cumulative nature of ordinal models contributes to more stable estimation procedures, particularly when sample sizes are moderate or when examining conditional relationships across multiple thresholds of the outcome variable.

Limitations

- **Consequences of Proportional Odds Violation -** When the proportional odds assumption is violated, the model's single odds ratio estimate represents a weighted average of the category-specific effects, potentially masking important heterogeneity in predictor effects across the outcome distribution.
- **Complexity in Handling Interaction Effects -** Interpreting interaction effects in ordinal logistic regression presents substantial challenges. The nonlinear nature of the model means that the interaction effect depends on the values of other covariates and the specific cumulative logit under consideration.
- **Sensitivity to Selection of Link Function -** Ordinal logistic regression's flexibility in accommodating various link functions such as logit, probit, complementary log-log. Different link functions can yield different conclusions, yet theoretical guidance for link selection is often limited.
- **Limited Software Capabilities and Diagnostic Tools -** Studies reveal a persistent gap in assumption testing, with limited reporting on proportional odds verification and model fit comparison in published research.

Ordinal logistic regression is the appropriate modeling choice when the outcome variable comprises three or more ordered categories, such as Likert scale responses (e.g., strongly disagree, disagree, neutral, agree, strongly agree), educational levels (e.g., high school, undergraduate, postgraduate), or disease severity stages (e.g., mild, moderate, severe).

This technique is specifically designed to preserve and utilize the natural ordering of the categories while estimating the probability of being at or below a particular threshold. Compared to multinomial logistic regression, which ignores the ordering and treats categories as unrelated, ordinal logistic regression is more parsimonious (estimating fewer parameters), more interpretable (providing a single coefficient per predictor across all thresholds), and statistically more powerful when the proportional odds assumption holds.

2.3.6 Support Vector Machines (SVM)

A Support Vector Machine (SVM) is a supervised learning model that classifies data by finding the optimal hyperplane which best separates different classes it was established by Cortes, C., & Vapnik, V. (1995). The fundamental concept is to maximize the margin, known as support vectors. For non-linearly separable data, SVM utilizes the kernel trick to map inputs into a higher-dimensional space, enabling it to find a clear dividing boundary without explicitly performing complex calculations.

Model

For the Support Vector Machine classification problem with *K* classes, let the training data be $(\boldsymbol{x_i}, \boldsymbol{y_i}), \textbf{where } \boldsymbol{x_i} \in \boldsymbol{R^p} \textbf{ and } \boldsymbol{y_i} \in \{\mathbf{1}, \mathbf{2}, \dots, \boldsymbol{K}\}$. In the direct multiclass (Crammer–Singer) formulation, we estimate a separate weight vector $\boldsymbol{w_k}$ and bias $\boldsymbol{b_k}$ for each class *k*. The decision function assigns a new observation *x* to the class with the largest score:

$$\hat{\boldsymbol{y}} = \textbf{arg}\max_{\boldsymbol{k}\in\{\mathbf{1},\dots,\boldsymbol{K}\}} \left(\boldsymbol{w_k^T x} + \boldsymbol{b_k}\right) \tag{2.11}$$

The optimization problem is formulated as:

$$\min_{\boldsymbol{w_k}, \boldsymbol{b_k}, \boldsymbol{\xi_i}} \frac{\mathbf{1}}{\mathbf{2}} \sum_{\boldsymbol{k}=\mathbf{1}}^{\boldsymbol{K}} |\boldsymbol{w_k}|^{\mathbf{2}} + \boldsymbol{C} \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{n}} \boldsymbol{\xi_i}$$

subject to the constraints:

$$\boldsymbol{w_{y_i}^\top x_i} + \boldsymbol{b_{y_i}} - (\boldsymbol{w_k^\top x_i} + \boldsymbol{b_k}) \geq \mathbf{1} - \boldsymbol{\xi_i}, \ \forall \boldsymbol{i}, \ \forall \boldsymbol{k} \neq \boldsymbol{y_i} \tag{2.12}$$

$$\boldsymbol{\xi_i} \geq \mathbf{0}.$$

Here, the constraints ensure that the score of the correct class exceeds the score of any other class by at least one unit (margin), allowing slack variables $\boldsymbol{\xi_i}$ for non-separable data. The regularization term $\frac{\mathbf{1}}{\mathbf{2}} \sum_{\boldsymbol{k}=\mathbf{1}}^{\boldsymbol{K}} |\boldsymbol{w_k}|^{\mathbf{2}}$ controls model complexity, and $C > 0$ balances margin maximization with classification errors. For nonlinear classification, inner products $\boldsymbol{x_i^T x_j}$ can be replaced by a kernel function $\boldsymbol{K}(\boldsymbol{x_i}, \boldsymbol{x_j})$, enabling the model to construct nonlinear decision boundaries while remaining linear in a higher-dimensional feature space.

Assumptions

- **Independent Observations:** SVM assumes that observations are independent and identically distributed (i.i.d.), meaning each data point is generated independently and does not influence others.
- **Margin-Based Separability:** SVM assumes that classes can be separated by a hyperplane with a reasonably large margin, either directly (linear case) or after transformation using a kernel (nonlinear case).
- **Correct Class Labels:** SVM assumes that class labels are accurate, because mis-labeled points especially near the boundary can strongly affect the position of the decision hyperplane.
- **Appropriate Kernel Choice:** For nonlinear classification, SVM assumes that a suitable kernel function exists that can map data into a space where classes become linearly separable.
- **No Distributional Assumptions:** SVM does not assume normality or any specific data distribution; it is a non-parametric, margin-based method focused on geometric separation.

Advantages

- **Strong Generalization Ability:** SVM maximizes the margin between classes, which helps reduce overfitting and improves performance on unseen data. This makes it highly effective in many real-world classification problems.
- **Effective in High-Dimensional Data:** SVM performs well when the number of features is large, even when the number of predictors exceeds the number of observations.
- **Handles Nonlinear Boundaries:** Using kernel functions (such as polynomial or radial basis function), SVM can model complex nonlinear decision boundaries without explicitly transforming the data.
- **Memory Efficient:** SVM relies only on support vectors (a subset of training data) to define the decision boundary, making it computationally efficient at prediction time.
- **No Distributional Assumptions:** SVM does not assume any specific probability distribution of predictors (such as normality), making it flexible and widely applicable across different types of data.

Limitations

- **Less Interpretable:** Compared to models like logistic regression, SVM does not provide easily interpretable coefficients or direct insights into how each predictor influences the outcome, particularly when nonlinear kernels are used.
- **No Direct Probability Output:** Standard SVM provides class labels but not probability estimates. Additional calibration methods are required if probability interpretation is needed.
- **Requires Proper Feature Scaling:** Since SVM relies on distance calculations and inner products, unscaled variables can distort results. Data pre-processing such as standardization is usually necessary.
- **Multiclass Extension is Indirect:** SVM was originally designed for binary classification. Multiclass classification requires extensions such as one-vs-one or one-vs-rest approaches, which increase computational complexity.

Support Vector Machine (SVM) classification is preferred when the dataset has a relatively high number of features compared to the number of observations and when strong generalization performance is required. It is particularly useful in high-dimensional problems such as text classification, image recognition, and bioinformatics, where clear class separation may exist in a transformed feature space.

SVM is also preferred when the decision boundary is complex and potentially nonlinear, as kernel functions allow flexible modeling without explicitly increasing dimensionality. Additionally, it is suitable when avoiding overfitting is important, since margin maximization improves robustness. However, it is most practical when the dataset size is moderate, properly scaled, and when computational cost is manageable.

2.3.7 Random Forest

The Random Forest algorithm was formally introduced by Breiman, L. (2001). It is an ensemble learning method that constructs a multitude of decision trees at training time and outputs the class that is the mode of the classes of the individual trees. It is designed to correct the tendency of individual decision trees to overfit their training data. The algorithm introduces two sources of randomness to achieve this. First, each tree is trained on a different bootstrap sample (a random sample with replacement) of the original data, a process known as bagging. Second, at each node split within a tree, only a random subset of the predictors is considered, this dual randomization effectively decorrelates the individual trees, making the ensemble robust against overfitting.

The final prediction is then aggregated across all trees, either by majority vote for classification or by averaging for regression, resulting in a model that often exhibits high predictive accuracy and the ability to capture complex, non-linear relationships.

Model

A random forest is a classifier consisting of a collection of tree-structured classifiers $\{\boldsymbol{h}(\mathbf{x}, \boldsymbol{\Theta_k}), \boldsymbol{k} = \mathbf{1}, \ldots\}$ where the $\{\boldsymbol{\Theta_k}\}$ are independent identically distributed random vectors and each tree casts a unit vote for the most popular class at input x.

this ensemble model consists of K randomized decision trees $\{\boldsymbol{h}(\mathbf{x}, \boldsymbol{\Theta}_1, \mathbf{D}_n), \boldsymbol{h}(\mathbf{x}, \boldsymbol{\Theta}_2, \mathbf{D}_n), \ldots, \boldsymbol{h}(\mathbf{x}, \boldsymbol{\Theta_k}\, \mathbf{D}_n)\}$, where each $\boldsymbol{\Theta_k}$ is an independent and identically distributed random vector governing tree construction. The trees are grown using the training data D_n with randomness introduced through mechanisms such as bootstrap sampling and random feature selection. For a new input vector x, the forest aggregates predictions through majority voting:

$$\hat{H}(\mathbf{x}) = \mathbf{arg}\max_{j} \sum_{k=1}^{K} \boldsymbol{I}(\mathbf{h}(\mathbf{x}, \boldsymbol{\Theta}_k, \mathbf{D}_n) = \mathbf{j}) \qquad (2.13)$$

$\boldsymbol{I}(.)$ is the indicator function.

The generalization error PE* = P(mg(X,Y) < 0) is bounded above by $\bar{\rho}(1-s^2)/s^2$.

Where, PE* is the Probability that forest misclassifies a new observation, mg(X,Y) is margin function, $\bar{\rho}$ is average correlation between trees and **s** is **E** {**X**, **Y**}[**mg**(**X**, **Y**)] is the average strength of individual trees.

Assumptions

- **Representative Data (I.I.D.):** The training data should be independent and identically distributed (i.i.d.) samples from the population to which results will be applied.
- **Independence of Randomness:** The random vectors (sampling with replacement and random feature subsets) generating each tree should be independent of past random vectors.
- **Sufficient Sample Size:** While Random Forest works in high-dimensional settings (p > n), some minimum sample size is needed for stable splits.
- **Low Correlation Between Trees:** The primary requirement for a strong random forest is that the individual decision trees are relatively uncorrelated with each other.
- **Predictive Power of Individual Trees:** Each tree in the forest must be better than random guessing (i.e., it must have a low error rate).

Advantages

- **Reduced Overfitting:** Unlike individual, deeply grown decision trees, Random Forests do not overfit as more trees are added. The generalization error converges as the number of trees increases.
- **Robustness to Noise and Outliers:** Because of the averaging/voting mechanism, RF is robust to outliers and noisy data.
- **Internal Validation (Out-of-Bag Error):** RF provides an "out-of-bag" (OOB) error estimate, which is an unbiased estimate of the test set error, removing the immediate need for a separate cross-validation set.
- **Handling High-Dimensional Data:** RF efficiently handles datasets with many features, even when the number of features exceeds the number of observations.
- **Versatility:** It can handle mixed data types (numerical and categorical) and requires minimal data pre-processing (e.g., no scaling or normalization needed).

Limitations

- **Limited Interpretability**: As an ensemble method, random forests sacrifice the transparency of individual decision trees. The aggregation of hundreds of trees creates a "black box" model, making it difficult to trace the rationale behind specific predictions.
- **Overfitting from Unrestricted Trees**: While increasing the number of trees does not cause overfitting, individual trees grown too deep can capture noise rather than signal. Without proper constraints on tree depth, the model becomes sensitive to minor fluctuations in the training data, compromising generalizability.
- **Compounded Sensitivity to Class Imbalance and Noise**: Classification accuracy degrades disproportionately when class imbalance and data noise co-occur.
- **Substantial Computational Demands**: The ensemble structure imposes significant computational costs, including large storage requirements, lengthy training times, and slow real-time prediction.

Random Forest is optimally employed when predictive accuracy in complex, nonlinear datasets is paramount and classical model assumptions are violated. It is particularly suitable when classical model assumptions such as linearity, normality, or independence among predictors are violated, since it is a non-parametric method.

Random Forest performs well in high-dimensional settings where the number of predictors is large, and it is robust to multicollinearity, outliers, and noise in the data. Additionally, it is advantageous when overfitting is a concern, as the ensemble aggregation of multiple decision trees reduces variance and improves generalization performance. It is also useful when internal model validation is needed, as it provides an out-of-bag error estimate without requiring a separate validation dataset.

Therefore, Random Forest is generally preferred for classification problems involving complex structures, mixed data types, and large datasets where prediction accuracy is prioritized over model interpretability.

2.3.8 XGBoost (Extreme Gradient Boosting)

XGBoost is an advanced ensemble machine learning algorithm based on gradient boosting that is widely used for classification problems it was developed by Chen, T., & Guestrin C. (2016). In classification, XGBoost builds a sequence of decision trees, where each new tree is trained to correct the errors made by the previous trees. Instead of building trees independently (like Random Forest), it builds them sequentially to minimize a loss function (such as logistic loss for binary classification or softmax for multi-class classification).

It uses gradient descent optimization to improve predictions and includes regularization to prevent overfitting, making it highly accurate and efficient. Conceptually, XGBoost combines many weak decision trees into a strong predictive model by continuously focusing on misclassified observations.

Model

In XGBoost for general classification, the model the prediction for observation $\boldsymbol{x_i}$ as an additive ensemble of *K* decision trees:

$$\widehat{\boldsymbol{y_i}} = \sum_{k=1}^{K} \boldsymbol{f_k}(\boldsymbol{x_i}), \quad \boldsymbol{f_k} \in \boldsymbol{\mathcal{F}} \tag{2.14}$$

where each $\boldsymbol{f_k}$ is a regression tree and $\boldsymbol{\mathcal{F}}$ denotes the space of possible trees. For multi-class classification with *C* classes, the model outputs a score vector:

$$\widehat{\boldsymbol{y_i}} = (\widehat{\boldsymbol{y_{i1}}}, \widehat{\boldsymbol{y_{i2}}}, \dots, \widehat{\boldsymbol{y_{iC}}})$$

These raw scores are converted into class probabilities using a suitable link function (typically softmax):

$$\boldsymbol{p_{ic}} = \frac{\exp(\widehat{y_{ic}})}{\sum_{j=1}^{c} \exp(\widehat{y_{ij}})} \tag{2.15}$$

The predicted class is:

$$\widehat{\boldsymbol{c_i}} = \arg\max_{c} \boldsymbol{p_{ic}}$$

Assumptions

- **Independence of Observations:** The training instances are assumed to be independently and identically distributed (i.i.d.). Correlated or dependent observations can reduce the reliability of the gradient updates and lead to overfitting or biased predictions.
- **Consistency of Feature-Target Relationship:** XGBoost assumes that patterns in the training data are representative of patterns in unseen data.
- **Additivity of Trees in Function Space:** The model assumes that the target function can be approximated as an additive combination of decision trees, where each successive tree corrects residual errors from the previous ensemble.
- **Differentiable Loss Function:** The algorithm requires that the chosen loss function for classification (e.g., logistic loss for binary classification or softmax cross-entropy for multi-class classification) is differentiable, as gradient and Hessian information is used to optimize tree splits.

Advantages

- **High Predictive Accuracy:** XGBoost consistently achieves superior predictive performance in classification tasks due to its gradient boosting framework, which iteratively minimizes residual errors and captures complex nonlinear relationships among predictors.
- **Flexibility with Data Types and Missing Values:** XGBoost can handle diverse predictor types, including numerical and categorical variables (after encoding), and can automatically manage missing data during tree construction. This reduces the need for extensive data pre-processing, increasing modelling efficiency.
- **Scalability and Computational Efficiency**: With parallelized tree construction, cache optimization, and support for out-of-core computing, XGBoost can efficiently handle large-scale classification problems, making it suitable for datasets with thousands of observations and features.
- **Interpretability through Feature Importance**: XGBoost provides multiple feature importance metrics, allowing researchers to identify the most influential predictors in classification outcomes.

Limitations

- **Sensitivity to Hyperparameters:** XGBoost requires careful tuning of multiple hyperparameters (e.g., learning rate, number of trees, tree depth). Improper tuning can lead to suboptimal performance or overfitting, and hyperparameter optimization can be computationally intensive.
- **Computational Resource Requirements:** Despite being optimized for efficiency, XGBoost can still demand significant computational resources, particularly for very large datasets or deep tree ensembles. Memory usage and training time may become limiting factors in resource-constrained environments.
- **Handling Highly Imbalanced Data:** In classification tasks with severe class imbalance, XGBoost may bias predictions toward majority classes if not properly adjusted with techniques such as class weighting, sampling strategies, or specialized evaluation metrics.
- **Limited Theoretical Guarantees:** While XGBoost often achieves high empirical performance, it lacks strong theoretical guarantees regarding convergence or statistical inference compared to traditional parametric models.

XGBoost is preferred in classification tasks when high predictive accuracy is a primary objective, particularly in datasets that are large, complex, or contain nonlinear relationships and interactions among predictors. Its ability to handle diverse data types, manage missing values automatically, and provide feature importance metrics makes it suitable for real-world applications where data preprocessing may be challenging.

Additionally, XGBoost is advantageous when computational efficiency is important, as it supports parallel processing and scalable learning on large datasets. Investigators and practitioners often choose XGBoost when traditional models, such as logistic regression or simple decision trees, fail to capture complex patterns, or when the dataset exhibits moderate to high dimensionality. Its robustness to overfitting through regularization and early stopping further enhances its applicability in empirical studies requiring reliable and generalizable classification performance.

2.3.9 Ordinal Forest

The Ordinal Forest method proposed by Hornung, R. (2020), is a tree-based ensemble method specifically designed to address the challenges of prediction and variable importance estimation when the target variable is ordinal in nature. It assigns optimized numerical scores to each ordinal category to maximize predictive performance, and it trains a standard regression Random Forest to predict these scores. This approach allows the model to inherently respect the natural order of the categories while automatically capturing complex, non-linear relationships and providing interpretable permutation-based variable importance measures.

Model

Let $(\mathbf{X_i}, \mathbf{Y_i}), \mathbf{i} = \mathbf{1}, \ldots, \mathbf{n}$, denote independent observations where $\mathbf{Xi} \in \boldsymbol{R^p}$ is a vector of predictors and $Y_i \in \{1,2,...,K\}$ is an ordinal response with natural ordering $1 < 2 < \cdots < K$. Instead of treating classes as purely nominal, the ordinal forest incorporates the ranking information by transforming the ordinal response into a set of binary cumulative indicators. For each bootstrap sample $b = 1,...,B$, a decision tree $\boldsymbol{T_b}$ is grown using recursive binary partitioning, where splits are selected to maximize an ordinal impurity reduction criterion (such as ordered Gini index or cumulative log-likelihood).

Each tree produces class probability estimates $\widehat{\boldsymbol{P_b}}(\boldsymbol{Y = k \mid X = x})$. The final prediction is obtained by aggregating across trees:

$$\widehat{\boldsymbol{P}}(\boldsymbol{Y = k \mid X = x}) = \frac{\mathbf{1}}{\boldsymbol{B}} \sum_{\boldsymbol{b=1}}^{\boldsymbol{B}} \widehat{\boldsymbol{P_b}}(\boldsymbol{Y = k \mid X = x}) \tag{2.16}$$

and the predicted class is

$$\widehat{\boldsymbol{Y}}(\boldsymbol{x}) = \mathbf{arg} \min_{\boldsymbol{k} \in \{\mathbf{1},\ldots,\boldsymbol{K}\}} \boldsymbol{E}\left[\boldsymbol{L}(\boldsymbol{Y}, \boldsymbol{k}) \mid \boldsymbol{X = x}\right] \tag{2.17}$$

where $\boldsymbol{L}(\boldsymbol{Y}, \boldsymbol{k})$ is an order-sensitive loss function (e.g., absolute error $|\boldsymbol{Y - k}|$). By incorporating order information into the splitting rule and aggregation step, the ordinal forest improves predictive performance and interpretability when modeling structured categorical outcomes compared to standard random forest classification.

Assumptions

- **Ordinal Response Structure**: The dependent variable consists of ordered categories ($1 < 2 < \cdots < K$), and this ranking is correctly specified.
- **Independence of Observations**: The data observations are independent and identically distributed (i.i.d.).
- **Sufficient Sample Size**: The dataset is large enough to construct multiple bootstrap samples and grow numerous trees for stable aggregation.
- **Order-Sensitive Splitting Criterion**: The ordinal nature of the outcome can be captured through recursive partitioning using an order-based impurity measure.
- **Reasonable Class Distribution**: Extreme class imbalance may affect probability estimation and prediction accuracy.
- **Ensemble Stability**; Aggregating a large number of decorrelated trees reduces variance and improves generalization.

Advantages

- **Proper Handling of Ordinal Structure:** The model explicitly incorporates the natural ordering of categories, ensuring that predictions respect rank information. This makes it more appropriate than nominal classifiers for ordered outcomes.
- **Reduced Misclassification Severity:** Ordinal Forest minimizes order-based loss rather than simple classification error.
- **No Strict Distributional Assumptions:** Ordinal Forest does not assume linearity, normality, or proportional odds. This flexibility is a major practical advantage in real-world data analysis.
- **Automatic Detection of Nonlinearities and Interactions:** The model can capture complex nonlinear relationships and interaction effects without requiring explicit specification. This enhances modelling power and predictive accuracy.
- **Improved Predictive Accuracy through Ensemble Learning:** By aggregating multiple decision trees (as in Random Forest), the model reduces variance and overfitting, resulting in better generalization performance.

Limitations

- **Limited Interpretability:** Although Ordinal Forest provides variable importance measures that ensemble structure makes it difficult to explain the exact relationship between predictors and the ordered outcome.
- **Dependency on Training Data Representation:** The performance of ordinal forests relies on the training data being a proper representation of the population. If the category distribution in the training set is not representative, predictions can be biased.
- **Sensitivity to Class Imbalance**: If the ordinal categories are highly imbalanced, the model may produce biased probability estimates and favor majority classes. Additional techniques such as resampling or class weighting may be required.
- **Lack of Explicit Probabilistic Model Structure**: Unlike parametric ordinal models (e.g., proportional odds models), Ordinal Forest does not provide an explicit mathematical equation describing the relationship between predictors and response. This limits theoretical interpretation and hypothesis testing.
- **Large Sample Requirement for Stability**: While tree-based models are flexible, reliable performance generally requires a sufficiently large sample size. With small datasets, predictions may become unstable despite ensemble averaging.

Ordinal Forest is preferred for classification when the response variable consists of ordered categories and the underlying relationship between predictors and the outcome is complex or unknown. It is particularly suitable when the ordinal structure of the dependent variable must be preserved and when misclassification severity depends on the distance between categories.

As an extension of Random Forest, it is especially useful when predictive accuracy is a primary objective and when the dataset is moderately large, allowing stable aggregation of multiple decision trees. Therefore, Ordinal Forest is generally preferred in applied classification problems where flexibility, robustness, and order-sensitive prediction are more important than strict interpretability of model parameters.

2.4 Confusion Matrix

A confusion matrix is a performance evaluation tool for classification models. It is nothing but an n × n matrix, where the value of n changes as per the number of the feature classes. It is a simple table that compares a model's predicted labels against the actual ground truth labels to visualize where the model is getting "confused".

For a binary classification problem (two classes, typically designated as "positive" and "negative"), the confusion matrix is a 2×2 table. By convention, the predicted classes are arranged in columns, and the actual classes are arranged in rows, although this orientation can sometimes be reversed.

Table 2.1 Confusion matrix

Actual / ***Predicted***	***Positives***	***Negatives***
Positives	True Positive (TP)	False Positive (FP)
Negatives	False Negative (FN)	True Negative (TN)

Where

- **True Positive (TP)** – True positive is the case when the actual and the predicted values are the same, which means that the model has rightly predicted the positive values.
- **True Negative (TN)** – False negative is the case when the actual and the predicted values are the same, which means that the model has rightly predicted the negative values.
- **False Positive (FP)** – False positive is the case when the classifier incorrectly predicts an actual negative value to be a positive value.
- **False Negative (FN)** – False negative is the case when the classifier incorrectly predicts an actual positive value to be a negative value.

Various performance measures can be derived using the confusion matrix. They are as follows:

Accuracy – Accuracy can be defined as the measure to calculate the number of right predictions that our classifier has made using the actual and the predicted values. Accuracy is mathematically represented as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision – Precision is the measure that evaluates the number of correctly predicted values that were positive. The reliability of the model can be deduced using precision.

$$Precision = \frac{TP}{TP + FP}$$

Recall/Sensitivity – Recall is the measure used to evaluate the number of actual positive values that our classifier was able to predict correctly among the predicted positive values. The recall is also known as the sensitivity of the model.

$$Recall = \frac{TP}{TP + FN}$$

Specificity – Specificity is also called the true-negative rate. It is the measure to calculate the number of rightly made negative predictions among the actual negative values.

$$Specificity = \frac{TN}{TN + FP}$$

F1 Score – F1 score is a metric that uses both precision and recall. It is the weighted average of both. F1 score can be calculated considering the harmonic mean of precision and Recall.

$$F1\ Score = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$

Experimental Study

CHAPTER III

EXPERIMENTAL STUDY

3.1 Introduction

The experimental evaluation of classification algorithms is a key component of this study, focusing on categorical response variables with different structures such as binary, multinomial, and ordinal. A major challenge in predictive modelling is not only selecting appropriate methods but also understanding how their performance varies across these different types of outcomes. This study compares both classical and modern classification techniques. Traditional methods such as Logistic Regression and Discriminant Analysis are evaluated alongside advanced machine learning approaches including Random Forest, Support Vector Machine, XGBoost, and Ordinal Forest. This combination allows for a comprehensive assessment of models with varying assumptions, complexity, and predictive capabilities.

Three case studies are considered, each representing a distinct classification problem. The models are tested under different conditions, including datasets with and without outliers, to examine robustness. Train-Test split validation is used to assess model generalizability, and key issues such as overfitting are also investigated.

Model performance is evaluated using accuracy by computing confusion matrices, providing insights into classification effectiveness. All analyses are conducted in R Software, with detailed results and code documented to ensure transparency and reproducibility.

This chapter presents the experimental study of applying various classification models to evaluate and compare their performance across binary, multinomial, and ordinal outcomes. It also examines the impact of outliers and the generalizability of models using Train–Test split validation. The data structure, as well as the method of outlier identification and removal used in this study, are explained in this chapter. All experimental results are generated using R programming, with the corresponding R syntax and detailed outputs provided in the appendices to ensure transparency and reproducibility.

3.2 Study Design

This study presents a comprehensive and structured investigation into the performance of various classical and machine learning classifiers across three fundamental prediction tasks binary, multinomial, and ordinal classification. The primary objective is to systematically evaluate model robustness, assess the impact of data outliers, and determine the generalizability of each algorithm to unseen data. By applying a consistent methodological framework to each case, this design enables a fair and meaningful comparison of classifier strengths and weaknesses under three experimental conditions.

The study is structured around three distinct case studies, each characterized by a different target variable type and dataset:

Case Study 1: Binary Classification Analysis - Heart Disease Prediction

Case Study 2: Multinomial Classification Analysis - Wine Class Prediction

Case Study 3: Ordinal Classification Analysis - BMI category prediction

Data Structure with nature of variable

The data structure assigned to a variable must accurately reflect its actual scale of measurement whether categorical ordered, unordered and numerical. When data is initially imported into R, the software automatically classifies each variable based on its syntax, typically treating it as either numerical (e.g., integer or double) or character (string). However, this default classification often fails to capture the true statistical nature of the variable.

As Jaynal Abedin (2014) emphasizes, understanding R object mode and classes is fundamental to effective data manipulation, to ensure valid statistical analysis. The necessary syntax for these structural modifications is provided in Appendix (R Syntax 1.1).

Identification and Removal of Outliers

Several methods are available for detecting and handling outliers in a dataset. Among these, the Inter Quartile Range (IQR) method is one of the most widely used due to its simplicity and robustness. In this study, outliers were identified and removed using the IQR method in R, and the corresponding syntax for this procedure is provided in Appendix (R Syntax 1.2).

Experimental conditions

This study investigates classifier performance across three fundamental outcome types such ass binary, multinomial, and ordinal with the goal of testing model robustness, evaluating the impact of outliers, and assessing generalizability to unseen data. To ensure consistency and comparability across all experiments, a standardized methodology is applied to each dataset and classification technique.

Each case studies performance evaluated under three experimental conditions:

- **Full Model (with outliers):** The complete dataset, including all observations, is used to train and evaluate each classifier. This condition assesses algorithmic robustness by examining how well models handle real-world data irregularities.

- **Full Model (without outliers):** Outliers are identified and removed prior to analysis, and the models are reevaluated on the cleaned dataset. This condition isolates the effect of outlier removal, revealing whether and to what extent extreme observations influence model performance and decision boundaries.

- **Train/Test Split (70/30):** To assess generalizability, each dataset is randomly split into 70% training and 30% testing sets. Models are trained on the training data and evaluated on the unseen test data. This process is repeated five times with different random splits, and accuracy scores are averaged to obtain stable out-of-sample performance estimates.

Case Study 1: Heart Disease Dataset – Binary Outcome

The binary classification experiment focuses on predicting the presence or absence of heart disease using a combination of clinical and demographic predictors. The dataset encompasses a range of risk factors and clinical measurements commonly associated with cardiovascular health. The binary outcome variable denotes whether heart disease is presence or absence, representing a classic binary classification problem in medical diagnostics.

Dataset Characteristics:

- **Objective:** Predict the presence or absence of heart disease.
- **Dataset:** Heart Disease dataset - Kaggle
- **Target:** Binary
- **Outcome:** (Absence, Presence)
- **Sample:** 735 observations (with outliers), 664 observations (without outlier).
- **Predictors:** Mix of continuous (e.g., Age, Cholesterol) and categorical (e.g., Gender).
- **Algorithms:**

    1) Binary Logistic Regression
    2) Random Forest
    3) Support Vector Machine (SVM)
    4) XGBoost.

Case Study 1: Binary Classification Analysis Results

This case study investigates the performance of four classification models such as Binary Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost on a binary outcome task predicting the presence of heart disease. The analysis evaluates model accuracy under three conditions: full data with outliers, full data without outliers, and Train/Test split validation. The objective is to assess not only predictive performance but also the generalizability and robustness of each model to outliers.

Table 1: Binary Prediction Model Accuracy under Conditions

	Full data (with outlier)	***Full data (without outlier)***	***Training/Testing Model***
Binary Logistic Regression	83.27%	82.53%	83.26%
Random Forest	100%	100%	82.44%
SVM	85.58%	85.54%	82.17%
XGBoost	93.61%	94.73%	80.36%

In the binary classification task, The above table shows that Random Forest achieved perfect accuracy comparatively other models on the full dataset, both with and without outliers, suggesting potential overfitting or the presence of easily separable classes in the data, as detailed in Appendix (R syntax 1.3) with one existing observation prediction.

However, the Train/Test split results provide a more realistic estimate of generalizability, with all models performing between 80-83%. Notably, Binary Logistic Regression showed the highest generalizable accuracy (83.26%), performing nearly as well as the more complex ensemble methods, as shown in Appendix (R syntax 1.4). The removal of outliers had a marginal impact on model performance.

Case Study 2: Wine Class Dataset – Multinomial Outcome

The multinomial classification experiment utilizes the classic Wine dataset, an inbuilt R dataset widely regarded as a benchmark for evaluating multiclass classification algorithms. This dataset contains chemical analyses of wines cultivated in the same region of Italy but derived from three different cultivars (Classes 1, 2, and 3), rendering it ideal for nominal outcome prediction where categories lack natural ordering.

Dataset Characteristics:

- **Objective:** Classify wines into one of three distinct cultivars. The categories have no natural order.
- **Dataset:** Classic R wine dataset
- **Target:** Multinomial (3 unordered wine cultivars).
- **Outcome:** Class 1,2 and 3
- **Sample:** 178 observations (with outliers), 161 observations (without outlier).
- **Predictors:** Continuous chemical measurements only (e.g., Alcohol, Flavonoids, Color Intensity).
- **Algorithms:**

    1) Quadratic Discriminant Analysis (QDA)
    2) Multinomial Logistic Regression
    3) Random Forest
    4) Support Vector Machine (SVM)
    5) XGBoost

Where the data follows a QDA assumptions such as predictors as multivariate normal distributions within each class and covariances of each class is unequal.

Case Study 2: Multinomial Classification Analysis Results

In this case study, multinomial classification techniques are applied to the Wine dataset, which consists of three distinct classes of wine cultivars. Models including Multinomial Logistic Regression, Quadratic Discriminant Analysis (QDA), Random Forest, SVM, and XGBoost are evaluated under the same three conditions. Special attention is given to the satisfaction of model assumptions, particularly for QDA, and how this influences predictive accuracy and generalizability.

Table 2: Multinomial Prediction Model Accuracy under Conditions

	Full data (with outlier)	***Full data (without outlier)***	***Training/Testing Model***
Multinomial Logistic Regression	100%	100%	91.48%
QDA	99.44%	100%	98.15%
Random Forest	100%	100%	97.04%
SVM	100%	100%	97.78%
XGBoost	100%	100%	94.07%

For the three-class wine dataset, The above table shows that nearly all models achieved perfect or near-perfect classification when evaluated on the full dataset both with and without outlier, especially the classical method multinomial logistic regression, which shows perfect accuracy as shown in Appendix (R syntax 1.5).

This strongly indicates that the wine classes are highly separable based on the provided features. The Train/Test split results offer greater insight, revealing that compared to other models QDA generalizes best with 98.15% accuracy as given in Appendix (R syntax 1.7). When model assumptions are met, the model can leverage its theoretical strengths to achieve greater predictive accuracy, as evidenced here.

To apply the QDA method, all predictor variables should be numerical in nature. This property is satisfied in this data, and the assumptions are verified to apply QDA, which is detailed in Appendix (R syntax 1.6).

Case Study 3: Heart Disease Dataset – Ordinal Outcome

The ordinal classification experiment addresses a particularly challenging and often underexplored problem in data analysis: predicting outcomes with naturally ordered categories. Using the Heart disease dataset, an ordinal outcome was constructed based on BMI categories, reflecting the progressive nature of cardiovascular disease risk with advancing BMI categories. This study emphasizes the importance of selecting appropriate models that can effectively capture the inherent ordering in the response variable.

Dataset Characteristics:

- **Objective:** Predict a patient's BMI category, a task that must respect the natural order of the categories.
- **Dataset:** Heart Disease dataset – Kaggle
- **Target:** Ordinal (4 ordered BMI category based on WHO guidelines).
- **Outcome:** Underweight < Normal < Overweight < Obese
- **Sample:** 735 observations (with outliers), 685 observations (without outlier).
- **Predictors:** Mix of continuous (e.g., Age, BP) and categorical (e.g., Gender, Active).
- **Algorithms:**

    1) Ordinal Logistic Regression
    2) Ordinal Forest
    3) Support Vector Machine (SVM)
    4) XGBoost

Case Study 3: Ordinal Classification Analysis Results

This case study addresses an ordinal classification problem involving the prediction of BMI categories ranging from underweight to obese. Given the ordered nature of the response, specialized models such as Ordinal Logistic Regression and Ordinal Forest are compared alongside SVM and XGBoost. The analysis highlights challenges related to class separability and overfitting, particularly in ensemble methods when applied to ordinal outcomes.

Table 3: Ordinal Prediction Model Accuracy under Conditions

	Full data (with outlier)	***Full data (without outlier)***	***Training/Testing Model***
Ordinal Logistic Regression	39.59%	39.12%	38.73%
Ordinal Forest	99.86%	95.33%	36.11%
SVM	51.16%	51.53%	37.19%
XGBoost	93.47%	93.14%	37.64%

In the ordinal classification task with BMI categories (underweight to obese), The above table results describes that Ordinal Logistic Regression demonstrated the most reliable performance with consistent accuracy around 39% across all data conditions, suggesting the proportional odds assumption holds.

Conversely, compared to other models Ordinal Forest achieved near-perfect accuracy (99.86%) on the full dataset but dropped to 36.11% in Train/Test split, indicating severe overfitting as shown in Appendix (R syntax 1.8) & Appendix (R syntax 1.9) respectively. The modest accuracy across all models (36-39%) suggests the BMI classes are not highly separable based on available features. Outlier removal had marginal impact, confirming the challenge lies in the predictive task itself rather than data quality.

Summary & Conclusions

CHAPTER IV

SUMMARY & CONCLUSIONS

4.1 Introduction

This experimental evaluation compares the performance of various classification algorithms across three distinct outcome types, namely binary, multinomial, and ordinal, with the objective of assessing predictive accuracy, generalizability, and robustness to outliers. The results provide several important insights into model selection, enhance the understanding of classification model performance, and highlight the critical role of aligning model assumptions with the underlying data characteristics.

This chapter presents the key findings, a comprehensive summary of the study, and discusses potential directions for future study in the subsequent sections.

4.2 Key Findings

1. **Model Performance is Task-Dependent:** No single algorithm consistently outperformed others across all tasks. While ensemble methods like Random Forest and XGBoost demonstrated high accuracy on full datasets, their performance often diminished in train/test scenarios, suggesting a propensity for overfitting. Simpler models, such as Binary Logistic Regression, proved to be more robust and generalizable in the binary task.

2. **Marginal Impact of Outlier Removal:** Across all three tasks, the removal of outliers had a minimal impact on model accuracy, typically resulting in changes of less than one percentage point. This suggests that the datasets are relatively robust to extreme values or that the outliers were not highly influential on the decision boundaries of the models tested.

3. **The Critical Role of Train/Test Validation:** Evaluating models on the full dataset consistently produced inflated accuracy estimates, sometimes reaching 100%. The train/test split provided a more realistic and conservative measure of generalizability, revealing performance drops that were otherwise obscured. This underscores the necessity of rigorous validation techniques to assess true predictive power.

4. **Assumptions Drive Accuracy in Parametric Models:** The multinomial case provided a clear demonstration of this principle. QDA achieved the highest generalizable accuracy (98.15%) because its underlying assumptions—multivariate normality and class-specific covariance are well-met by the wine dataset. When theoretical assumptions hold, parametric models can outperform more flexible, assumption-free algorithms.

5. **Ordinal Structure Requires Specialized Methods:** The ordinal classification task highlighted the inadequacy of standard classifiers for ordered outcomes. Models like SVM and XGBoost, which treat the target variable as nominal and ignore its inherent order, The consistently low accuracy (36-39%) achieved by all models in the BMI category prediction task highlights a fundamental challenge in ordinal classification that cannot be solved by simply applying more complex algorithms. While Ordinal Logistic Regression provided the most stable performance by satisfying the proportional odds assumption, its modest accuracy suggests that the assumption alone is insufficient when classes are not well-separated based on available features.

4.3 Summary

In summary, modern machine learning algorithms (Random Forest, XGBoost, SVM) achieved high accuracy across all tasks but do not consider statistical assumptions. However, when theoretical assumptions were satisfied, classical methods excelled: QDA achieved superior generalization (98.15%) in the multinomial task, simpler logistic regression generalized best for binary outcomes, and for ordinal prediction modern methods like OrdinalForest can achieve perfect accuracy on full data, their performance drops sharply when making predictions on new data, revealing that ordinal prediction tasks need models that are not only flexible but also capable of generalizing well to unseen cases.

This underscores that while modern methods provide robust baselines, classical models can outperform them when their underlying assumptions are met.

4.4 FUTURE SCOPE

The experimental results clearly show that ordinal prediction tasks, like classifying BMI categories, present unique challenges that current models struggle to handle effectively. While modern methods like OrdinalForest can achieve perfect accuracy on training data, they fail to generalize, dropping to just 36% on new predictions. This reveals a key problem: we need models that are not only flexible but also capable of learning patterns that work on unseen data.

Practical Suggestions for Improvement:

- **Combine Classical Assumptions with Modern Flexibility:** Instead of choosing between traditional and modern methods, develop hybrid approaches that respect the natural order of categories while capturing complex patterns. For example, use ordinal logistic regression as a base and add flexibility through splines or nonlinear terms.
- **Use Ordinal-Specific Evaluation Metrics:** Standard accuracy can be misleading for ordinal data. Use metrics that penalize predictions based on how far they are from the true category (e.g., predicting "obese" when the true class is "overweight" should be treated as a smaller error than predicting "underweight").
- **Apply Regularization Techniques:** To prevent the severe overfitting seen with OrdinalForest, incorporate regularization methods that constrain model complexity while preserving the ability to detect meaningful patterns.
- **Consider Cumulative Probability Models**: Instead of predicting a single category, model the probability of being in each category or below. This approach naturally respects the ordinal structure and can be combined with machine learning techniques.
- **Collect More Discriminative Data:** If possible, gather additional predictors that better capture the underlying continuum that ordinal categories represent. The modest accuracy across all models suggests the current features simply may not contain enough information to reliably separate the categories.

Ordinal-based classification requires dedicated research to develop models that truly respect category order while maintaining both flexibility and the ability to generalize well to new data.

Appendix

APPENDIX

syntax 1.1

Data structure when data imported in R

```
> ############Data##############
>
> library(readxl)
> data = read_excel(file.choose(),sheet=1)
> str(data)
tibble [735 × 11] (S3: tbl_df/tbl/data.frame)
 $ Cholesterol   : num [1:735] 160 236 117 255 220 254 209 197 216 226 ...
 $ Age           : num [1:735] 39.2 39.4 39.4 39.4 39.5 ...
 $ gender        : chr [1:735] "Female" "Male" "Male" "Male" ...
 $ Active        : chr [1:735] "Yes" "Yes" "Yes" "Yes" ...
 $ BMI           : num [1:735] 25.1 23.2 24.1 23.5 23.2 ...
 $ S_BP          : num [1:735] 110 120 120 120 110 120 130 160 120 123 ...
 $ MaxHR         : num [1:735] 185 144 134 161 144 110 178 152 116 127 ...
 $ ExerciseAngina: chr [1:735] "No" "No" "Yes" "Yes" ...
 $ Oldpeak       : num [1:735] 0 0.1 2.5 0 0.4 0 0 1.2 1.8 1.7 ...
 $ ST_Slope      : chr [1:735] "Downsloping" "Downsloping" "Upsloping" "Downsloping" ...
 $ HeartDisease  : chr [1:735] "Absence" "Absence" "Presence" "Presence" ...
```

Data structure as per the variable nature

```
> #After changed the data structure
>
> data$gender = factor(data$gender)
> data$Active = factor(data$Active)
> data$ExerciseAngina = factor(data$ExerciseAngina)
> data$HeartDisease = factor(data$HeartDisease)
>
> # Convert ST_Slope to ordered factor
> data$ST_Slope <- factor(
+   data$ST_Slope,
+   levels = c("Upsloping", "Flat", "Downsloping"),
+   ordered = TRUE)
>
> str(data)
tibble [735 × 11] (S3: tbl_df/tbl/data.frame)
 $ Cholesterol   : num [1:735] 160 236 117 255 220 254 209 197 216 226 ...
 $ Age           : num [1:735] 39.2 39.4 39.4 39.4 39.5 ...
 $ gender        : Factor w/ 2 levels "Female","Male": 1 2 2 2 2 1 2 2 2 1 ...
 $ Active        : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 1 1 2 ...
 $ BMI           : num [1:735] 25.1 23.2 24.1 23.5 23.2 ...
 $ S_BP          : num [1:735] 110 120 120 120 110 120 130 160 120 123 ...
 $ MaxHR         : num [1:735] 185 144 134 161 144 110 178 152 116 127 ...
 $ ExerciseAngina: Factor w/ 2 levels "No","Yes": 1 1 2 2 1 1 1 1 2 2 ...
 $ Oldpeak       : num [1:735] 0 0.1 2.5 0 0.4 0 0 1.2 1.8 1.7 ...
 $ ST_Slope      : Ord.factor w/ 3 levels "Upsloping"<"Flat"<..: 3 3 1 3 1 3 3 $
 $ HeartDisease  : Factor w/ 2 levels "Absence","Presence": 1 1 2 2 1 1 1 1 2 2$
>
```

syntax 1.2

R Syntax for outlier removal

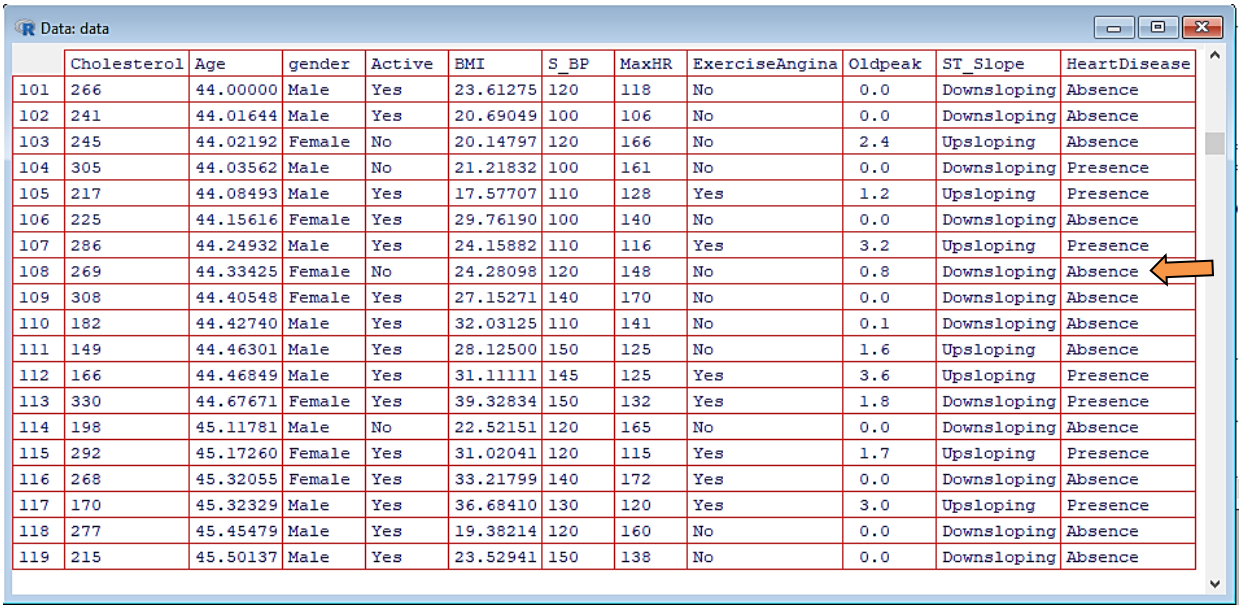
```
> # Import the data
>
> # to get dimension of data, no. of Rows * Columns
>
> dim(data)
[1] 735  11
>
> # R Code to remove Outliers
>
> remove_outliers_full <- function(df) {
+   num_cols <- sapply(df, is.numeric)
+
+   for (col in names(df)[num_cols]) {
+     Q1 <- quantile(df[[col]], 0.25, na.rm = TRUE)
+     Q3 <- quantile(df[[col]], 0.75, na.rm = TRUE)
+     IQR <- Q3 - Q1
+
+     lower <- Q1 - 1.5 * IQR
+     upper <- Q3 + 1.5 * IQR
+
+     df <- df[df[[col]] >= lower & df[[col]] <= upper, ]
+   }
+   return(df)
+ }
>
> clean_data <- remove_outliers_full(data)
>
> dim(clean_data)
[1] 664  11
>
```

syntax 1.3

Random Forest model in Binary Case (full data with outliers)

```
> #############Packages required############
>
> #install.packages("randomForest") # For Random Forest
> #library(randomForest)
>
> #random Forest (Full Data)
>
> # Fit Random Forest model
> rf_model <- randomForest(
+   HeartDisease ~ . ,
+   data = data,
+   ntree = 500,
+   mtry = 3,
+   importance = TRUE)
>
> # Predicted class
> rf_pred <- predict(rf_model, newdata = data)
>
> # Confusion matrix
> rf_conf_matrix <- table(Predicted = rf_pred,
+                         Actual = data$HeartDisease)
> rf_conf_matrix
          Actual
Predicted  Absence Presence
  Absence      387        0
  Presence       0      348
>
> # Accuracy
> rf_f_accuracy <- sum(diag(rf_conf_matrix))*100 / sum(rf_conf_matrix)
> rf_f_accuracy
[1] 100
>
```

To predict a single observation (108) in Existing heart disease dataset



Data: data

	Cholesterol	Age	gender	Active	BMI	S_BP	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
101	266	44.00000	Male	Yes	23.61275	120	118	No	0.0	Downsloping	Absence
102	241	44.01644	Male	Yes	20.69049	100	106	No	0.0	Downsloping	Absence
103	245	44.02192	Female	No	20.14797	120	166	No	2.4	Upsloping	Absence
104	305	44.03562	Male	No	21.21832	100	161	No	0.0	Downsloping	Presence
105	217	44.08493	Male	Yes	17.57707	110	128	Yes	1.2	Upsloping	Presence
106	225	44.15616	Female	Yes	29.76190	100	140	No	0.0	Downsloping	Absence
107	286	44.24932	Male	Yes	24.15882	110	116	Yes	3.2	Upsloping	Presence
108	269	44.33425	Female	No	24.28098	120	148	No	0.8	Downsloping	Absence
109	308	44.40548	Female	Yes	27.15271	140	170	No	0.0	Downsloping	Absence
110	182	44.42740	Male	Yes	32.03125	110	141	No	0.1	Downsloping	Absence
111	149	44.46301	Male	Yes	28.12500	150	125	No	1.6	Upsloping	Absence
112	166	44.46849	Male	Yes	31.11111	145	125	Yes	3.6	Upsloping	Presence
113	330	44.67671	Female	Yes	39.32834	150	132	Yes	1.8	Downsloping	Presence
114	198	45.11781	Male	No	22.52151	120	165	No	0.0	Downsloping	Absence
115	292	45.17260	Female	Yes	31.02041	120	115	Yes	1.7	Upsloping	Presence
116	268	45.32055	Female	Yes	33.21799	140	172	Yes	0.0	Downsloping	Absence
117	170	45.32329	Male	Yes	36.68410	130	120	Yes	3.0	Upsloping	Presence
118	277	45.45479	Male	Yes	19.38214	120	160	No	0.0	Downsloping	Absence
119	215	45.50137	Male	Yes	23.52941	150	138	No	0.0	Downsloping	Absence

Syntax to predict one of the observations in dataset

```
> #To verify RandomForest model Predict with 100% Accuracy (in Binary with outlier)
>
> # Patient Vicky
>
> # Select the first row as an example
> single_observation <- data[108, ]  # This will have all the correct factor structures
>
> # Make prediction
> rf_pred_v <- predict(rf_model, newdata = single_observation)
> print(rf_pred_v)
       1
Absence
Levels: Absence Presence
```

The above result shows predicting accuracy of Random Forest in binary case.

syntax 1.4

Binary Logistic regression model (Training [70%] Testing [30%] split)

```
> ########################################################
> #Binary Logistic Regression (Training testing model)
>
> train_index <- sample(1:nrow(data), 0.70 * nrow(data))
> train_data <- data[train_index, ]
> test_data  <- data[-train_index, ]
>
>
> bl_model <- glm(HeartDisease ~ . ,
+               data = train_data,
+               family = binomial(link = "logit"))
>
> # Predicted probabilities
> test_prob <- predict(bl_model,
+                      newdata = test_data,
+                      type = "response")
>
> pred_class <- ifelse(test_prob >= 0.5, 1, 0)
>
> # Confusion matrix
> bl_tt_conf_matrix <- table(Predicted = pred_class,
+                      Actual = test_data$HeartDisease)
> bl_tt_conf_matrix
         Actual
Predicted Absence Presence
        0      96       19
        1      18       88
> # Accuracy
> bl_tt_accuracy <- sum(diag(bl_tt_conf_matrix)) / sum(bl_tt_conf_matrix)
> bl_tt_accuracy
[1] 0.8325792
>
```

syntax 1.5

Multinomial Logistic regression model (full data without outliers)

```
> #############Packages required############
>
> #install.packages("nnet")  #for Multinomial logistic reg
> library(nnet)
>
> #Multiple Logistic Regression (Full Data without outlier)
>
> # After removing outliers data named as "clean_data"
>
> wo_ml_model <- multinom(Class ~ ., data = clean_data)
> # Predicted probabilities
> wo_test_prob <- predict(wo_ml_model,
+                         newdata = clean_data,
+                         type = "probs")
>
> # Convert probability matrix to predicted class
> wo_pred_class <- colnames(wo_test_prob)[apply(wo_test_prob, 1, which.max)]
>
> # Convert to factor (match original outcome levels)
> wo_pred_class <- factor(wo_pred_class, levels = levels(clean_data$Class))
>
> # Confusion matrix
> wo_conf_matrix <- table(Predicted = wo_pred_class,
+                         Actual = clean_data$Class)
> wo_conf_matrix
         Actual
Predicted  1  2  3
        1 57  0  0
        2  0 61  0
        3  0  0 43
>
> # Accuracy
> ml_wo_accuracy <- sum(diag(wo_conf_matrix))*100 / sum(wo_conf_matrix)
> ml_wo_accuracy
[1] 100
>
```

syntax 1.6

```
#Discriminant Analysis (LDA/QDA) Assumptions verifying R code given below

############################
#Assumptions

#1. Normality test
install.packages("MVN")
library(MVN)

# For each class separately - using subset()
mvn(data = subset(data, Class == levels(data$Class)[1], select = -Class))

mvn(data = subset(data, Class == levels(data$Class)[2], select = -Class))

mvn(data = subset(data, Class == levels(data$Class)[3], select = -Class))


#2.Equality of covariance matrices

install.packages("biotools")
library(biotools)

boxM(data[, -1], data$Class)



#3. Multicollinearity among predictors

#a) Correlation matrix

cor(data[, -1])


#4.Outliers

library(MASS)

check_outliers <- function(df) {
  md <- mahalanobis(df, colMeans(df), cov(df))
  cutoff <- qchisq(0.975, df = ncol(df))
  which(md > cutoff)
}

# Correct bracket notation - exclude first column properly
check_outliers(data[data$Class == levels(data$Class)[1], -1])

check_outliers(data[data$Class == levels(data$Class)[2], -1])

check_outliers(data[data$Class == levels(data$Class)[3], -1])

############################
```

syntax 1.7

QDA model in Multinomial case (Training [70%] Testing [30%] split)

```
> ################################################
> #QDA using Training Testing splits
>
> # Convert to matrix
> library(MASS)
>
> predictor_vars <- setdiff(names(data), "Class")
>
> # Split data
> train_index <- sample(1:nrow(data), 0.70 * nrow(data))
> train_data <- data[train_index, ]
> test_data  <- data[-train_index, ]
>
> # Fit QDA on training data
>
> tt_qda_model <- qda(Class ~ ., data = train_data)
>
> levels(train_data$Class)
[1] "1" "2" "3"
>
> tt_qda_pred <- predict(tt_qda_model, newdata = test_data)
>
> tt_qda_class <- tt_qda_pred$class
>
> tt_qda_cm <- table(Predicted = tt_qda_class, Actual = test_data$Class)
>
> qda_tt_accuracy <- sum(diag(tt_qda_cm))*100 / sum(tt_qda_cm)
>
> tt_qda_cm
         Actual
Predicted  1  2  3
        1 22  0  0
        2  0 18  0
        3  0  0 14
> qda_tt_accuracy
[1] 100
```

syntax 1.8

Ordinal Forest model (full data with outliers)

```
> #Ordinal Forest Regression
>
> #OrdinalForest(Full model)
>
> # 1. Ensure data is a standard data frame
> data_df <- as.data.frame(data)
>
> # 2. Ensure the dependent variable is an ordered factor
> data_df$BMI_Group <- factor(data_df$BMI_Group, ordered = TRUE)
>
> # 3. Run the model using the cleaned data frame
> ordRF <- ordfor(depvar = "BMI_Group",
+                 data = data_df,
+                 nsets = 100,
+                 ntreeperdiv = 100,
+                 ntreefinal = 500)
>
> f_preds <- predict(ordRF, newdata = data_df)
>
> # Extract the predicted class labels
> f_pred_class <- f_preds$ypred
>
> f_conf_matrix <- table(Actual = data_df$BMI_Group, Predicted = f_pred_class)
> print(f_conf_matrix)
             Predicted
Actual        Underweight Normal Overweight Obese
  Underweight           8      0          0     0
  Normal                0    257          0     0
  Overweight            0      0        280     1
  Obese                 0      0          0   189
>
> # Calculate Accuracy manually
> of_f_accuracy <- sum(diag(f_conf_matrix))*100 / sum(f_conf_matrix)
> of_f_accuracy
[1] 99.86395
>
```

syntax 1.9

Ordinal Forest model (Training [70%] Testing [30%] split)

```
> #Ordinal forest (with traing testing model)
>
> data_df <- as.data.frame(data)
>
> data_df$BMI_Group <- factor(data_df$BMI_Group, ordered = TRUE)
>
>
> train_index <- sample(seq_len(nrow(data_df)),
+                       size = 0.70 * nrow(data_df))
>
> train_data <- data_df[train_index, ]
> test_data  <- data_df[-train_index, ]
>
> tt_ordRF <- ordfor(depvar = "BMI_Group",
+                 data = train_data,
+                 nsets = 100,
+                 ntreeperdiv = 100,
+                 ntreefinal = 500)
>
> tt_preds <- predict(tt_ordRF, newdata = test_data)
>
> # Extract the predicted class labels
> tt_pred_class <- tt_preds$ypred
>
> tt_pred <- factor(tt_pred_class,
+                   levels = levels(data$BMI_Group),
+                   ordered = TRUE)
>
> tt_conf_matrix <- table(Actual = test_data$BMI_Group, Predicted = tt_pred)
>
> # Calculate Accuracy manually
> of_tt_accuracy <- sum(diag(tt_conf_matrix)) / sum(tt_conf_matrix)
> of_tt_accuracy
[1] 0.3484163
>
```

References

References

- Agresti, A. (2013). Categorical Data Analysis (3rd ed.). Wiley, New York.
- Breiman, L. (2001). Random Forest Regressions. Machine Learning, 45(1), 5-32.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system, Proceedings of the 22nd ACM SIGKDD International Conf. on Knowledge Discovery and Data Mining, 785–794.
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273–297.
- Cox, D. R. (1958). The regression analysis of binary sequences. Journal of the Royal Statistical Society. Series B (Methodological), 20(2), 215-242.
- Dua, D., & Graff, C. (2019). UCI Machine Learning Repository (Wine Dataset). University of California, Irvine.
- Fisher, R. A. (1936). The Use of Multiple Measurements in Taxonomic Problems. Annals of Eugenics, 7, 179-188.
- Hornung, R. (2020). Ordinal Forests. Journal of Classification, 37(1), 4–17.
- Hosmer, D. W., Jr., Lemeshow, S., & Sturdivant, R. X. (2013). Applied Logistic Regression (3rd ed.). John Wiley & Sons, New York.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning: With Applications in R (2nd ed.). Springer, New York.
- Jaynal Abedin (2014). Data manipulation with R. Packt Publishing, UK.
- Martinovic, M. (2025). Comparative Analysis of Machine Learning Models for Performance Evaluation Using Repeated k-Fold Cross-Validation. Applied Sciences, 15(7), 3636.
- McCullagh, P. (1980). Regression models for ordinal data. Journal of the Royal Statistical Society: Series B (Methodological), 42(2), 109–142.
- McCullagh, P., & Nelder, J. A. (1989). Generalized Linear Models (2nd ed.). Chapman & Hall.

- Muthukrishnan, R & Maryam Jamila, S. (2020). Predictive Modeling Using Support Vector Regression. International Journal of Scientific & Technology Research, 9 (2), 4863-4865.
- R Core Team (2025). R: A Language and Environment for Statistical. Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Smith, C. A. B. (1947). Some examples of discrimination. Annals of Eugenics, 13, 272-282.
- Stevens, S. S. (1946). On the theory of scales of measurement. Science, 103(2684), 677–680.
- Theil, H. (1969). A multinomial extension of the linear logit model. International Economic Review, 10(3), 251–259.
- Yu, B. (2012). Analysis of a Random Forests Model. Journal of Machine Learning Research, 13, 1063–1095.

❋ ❋ ❋